BAYESIAN INFERENCE FOR MARKOV KERNELS VALUED IN WASSERSTEIN SPACES

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Abstract. In this work, the author analyzes quantitative and structural aspects of Bayesian inference using Markov kernels, Wasserstein metrics, and Kantorovich monads. In particular, the author shows the following main results: first, that Markov kernels can be viewed as Borel measurable maps with values in a Wasserstein space; second, that the Disintegration Theorem can be interpreted as a literal equality of integrals using an original theory of integration for Markov kernels; third, that the Kantorovich monad can be defined for Wasserstein metrics of any order; and finally, that, under certain assumptions, a generalized Bayes's Law for Markov kernels provably leads to convergence of the expected posterior distribution in the Wasserstein metric to an optimal value. These contributions provide a basis for studying further convergence, approximation, and stability properties of Bayesian inverse maps and inference processes using a unified theoretical framework that bridges between statistical inference, machine learning, and probabilistic programming semantics.

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1. Introduction

Bayesian inference is fundamental to data-driven mathematical, statistical, and computational research. Many fields that might traditionally be called deterministic have, for some time now, embraced a Bayesian perspective. In the field of inverse problems, for example, Bayesian methods date back to at least the late sixties [62], and they have since become as fundamental as optimization-based approaches; for an introduction and overview, see, for example, [23] or [32]. Bayesian methods have similarly begun to proliferate in numerical analysis under the name of probabilistic numerics; see [18] for a recent review.

Roughly, the fundamental question guiding these developments might be distilled to this: How do we better incorporate information and uncertainty into our models? Much of modern statistical inference and machine learning goes to a further extreme and asks: How might we eliminate models altogether? However, even in the most extreme cases of model-free thinking—take, for example, generative adversarial networks (GANs), first defined by Goodfellow et al. in [30]—Bayesian interpretations still abound. In [47], for example, the authors argue that ordinary GANs can be viewed as a special case of a more general Bayesian GAN, and in [56], the authors argue that cycle-consistent, or bidirectional, adversarial learning, wherein one attempts to learn both generative and inferential models, should be viewed as a form of approximate Bayesian inference.

Rather than think in terms of model-based versus model-free methods, we might instead say that both are subject to underlying laws of information dynamics. In either case, the goal is to acquire accurate, reliable information about systems of interest, and Bayesian inference is arguably a common calculus, or, at least, one possible common calculus, implicit in any given mixture of methods.

More recently, probabilistic programming languages have emerged as a promising paradigm for combining the best of model-based and model-free thinking; see [58] for an introduction and overview. Two popular languages are Pyro [8], implemented in Python, and Gen [20], implemented in Julia. In a probabilistic programming language, probability distributions are taken as fundamental types that can be directly manipulated and transformed via programs. Parts of such programs may look more like the kind of code common to scientific computing, albeit with probabilistic uncertainty built directly into the variables. Other parts of the program may incorporate neural networks where models are unavailable and everything must be approximated purely from data. After specifying such a program, inference (and learning) is implemented automatically underneath the hood without the need for additional, special-purpose model-fitting algorithms.

In addition to the work being done on implementing such languages, much has been done to investigate their underlying theoretical and semantic properties; see, for example, [6], [54], and [55]. Markov kernels and monads figure prominently in these works, as they provide a convenient abstraction for the transformations implemented by the programs in a probabilistic programming language. Additionally, Markov kernels provide an elegant language for defining Bayesian inverse maps precisely. However, this work also tends to be purely measure-theoretic. As such, while

these approaches are structurally elegant, they do not immediately connect with quantitative questions in statistical inference and probabilistically-grounded areas of machine learning, where things like convergence, stability, and approximation results are key.

Partly to bridge this divide, and partly to simply better understand the mathematical foundations of Bayesian inference in a general setting, the present work establishes a theoretical framework for analyzing Bayesian inference processes using Markov kernels valued in Wasserstein spaces. By placing metrics on the domain and range of a Markov kernel, and by further viewing collections of Wasserstein-valued Markov kernels as something like generalized L^p-spaces, we arrive at a theory that is simultaneously structural and quantitative. That is, we retain the monadic, Markov-kernel-based framework of various semantic theories while also gaining the ability to analyze convergence, approximation, and stability properties of ordinary statistical inference problems. Moreover, such a framework can also be used to analyze, at a high level and without reference to particular implementations via neural networks, the inferential processes present in GANs and other, similar learning methods.

1.1. **Related Work.** Our work is closely related to two different lines of research that, until now, have mostly been pursued separately. First, many of the structural ideas in our approach come directly from research in categorical probability theory and probabilistic programming semantics. The papers [14, 17, 19, 22, 50] and [51] all develop categorical approaches to Bayesian inference. Our characterization of Bayesian inverse maps as certain kinds of Markov kernels comes, in particular, from [17] and [22], though our formulation extends theirs by including metric information. All of these papers ultimately build on a categorical approach to probability theory using Markov kernels and monads first developed by Lawvere [37] in an unpublished appendix and later formalized and popularized by Giry [29]. Independently, Cencov (also transliterated as Chentsov) defined and studied a category equivalent to the category of Markov kernels [12]; (this paper is only available in Russian, but his comprehensive monograph encompassing such work is available in translation [13].)

The above papers are all primarily measure-theoretic, which is to say: They offer much structural insight into probabilistic and inferential concepts, but they are ill-suited for quantitative investigations of convergence, stability, and approximation. The Kantorovich monad [27, 28, 57] is an exception, since it contains metric information, but, to our knowledge, it has not been used in a study of Bayesian inference.

Our work is connected, to a greater degree, with statistical investigations of optimal transport and with applications of optimal transport to machine learning. Most papers that specifically study Bayesian inference using optimal-transport based methods do so via Wasserstein gradient flows [2], and the focus is primarily on designing algorithms; see, for example, [1, 7, 39, 61]. Wasserstein barycenters have also been used for approximate Bayesian inference, and the work here is closer to our own; recent papers here include [4, 5, 53]. The Wasserstein metric is also increasingly popular in statistical learning as an alternative to KL-divergence. Papers here include [38, 44], and [49]. In particular, Singh and Póczos [49] investigate minimax

rates for estimating a probability distribution under a Wasserstein loss, and in their proofs, they appeal exclusively to the metric properties of the Wasserstein distance. Our own proofs frequently rely on similar partitioning arguments and further confirm the viability of developing statistical results using metric measure spaces, as opposed to Euclidean or Hilbert spaces, as an underlying foundation.

Finally, we make special mention of three papers, in particular, that we have found influential. First, Patterson [45] uses a mix of categorical and optimal-transport based methods to define Hausdorff and Wasserstein metrics for structured data. What we are calling the Markov-Wasserstein metric is due to him, and it plays an important part in our own work. Additionally, Ronan [46] investigates interrelationships between Markov kernels and optimal transport and shows that this machinery can extend optimal transport to the case of signed measures. Lastly, perhaps closest to our own work is the paper [31] by Jost et al., which contains a generalized Bayes's Law for highly general probabilistic morphisms. In Theorem 5.4, we prove a similar result using simpler methods, albeit in a more restricted setting.

1.2. **Outline.** In Section 2, we introduce Wasserstein spaces and Markov kernels. The material on Wasserstein spaces is standard, whereas the material on Markov kernels diverges somewhat from other presentations. Since Bayesian inverse maps are, ultimately, a special type of disintegration, we present, in Section 3, an original theory of integration for Markov kernels that leads to a modified version of the Disintegration Theorem, namely Theorem 3.16, which has a satisfying conceptual interpretation. Then, in Section 4, we show that the Kantorovich monad can be defined for Wasserstein metrics of any order. In Section 5, we show how these components come together in the setting of Bayesian inference. First, we define Bayesian inverse maps using Theorem 3.16. Then, we use our approximation machinery from Section 3 to give a simplified proof of a generalized Bayes's Law in Theorem 5.4. Finally, we show two convergence results for the expected Bayesian update in Proposition 5.7 and Theorem 5.10.

2. Wasserstein Spaces and Markov Kernels

In this section, we introduce a litany of definitions and foundational results on Wasserstein spaces and Markov kernels. The material on Wasserstein spaces is, for the most part, standard and should be background for anyone familiar with Villani's books, [59] or [60]. We do, however, take a slightly non-standard approach to couplings by viewing them as elements of a Wasserstein space in their own right. We explain this difference of perspective in Proposition 2.11 and the surrounding discussion.

The material on Markov kernels is more original. It is well-known that an ordinary Markov kernel can be viewed as a measurable, distribution-valued function, and the proof of equivalence is straightforward; see Lemma 3.1 in [33]. In our setting, we likewise want to view a Markov kernel as a Borel map with values in a Wasserstein space. However, the proof from the ordinary setting does not immediately apply, and in fact, substantially more work is required. In Theorem 2.25, we verify that the desired equivalence does hold.

2.1. **Wasserstein Spaces.** First, we present a number of definitions and results that we will use throughout the rest of the paper. We work in the setting of Polish metric spaces, since this guarantees that disintegrations of joint probability distributions exist. Additionally, it is a remarkable fact that the Wasserstein space associated to a Polish metric space is itself Polish. As will be seen, we can therefore iterate the Wasserstein construction and study, among other things, distributions on spaces of distributions.

Definition 2.1. A **Polish space** is a separable completely metrizable topological space X. That is, X is homeomorphic to a complete metric space (Y, d_Y) that has a countable dense subset. We sometimes call the pair (Y, d_Y) Polish as shorthand for a complete separable metric space.

Definition 2.2. A measure space (X, Σ_X) is called **standard Borel** if there exists a complete separable metric d on X such that Σ_X is equal to the Borel σ-algebra Σ_d induced by d. Note that if (X, Σ_X) is standard Borel, then X is Polish.

Definition 2.3. A **metric measure space** (X, d_X, μ_X) is a complete separable metric space (X, d_X) equipped with a σ-finite measure μ_X on Σ_{d_X}. We typically denote Σ_{d_X} by Σ_X.

Definition 2.4. Let (X, d_X) be Polish, and let $1 \leq p < \infty$. A Borel probability measure μ on (X, Σ_X) has **finite pth moment** if

$$\int_X d_X^p(x_0, x)\mu(dx) < \infty$$

for some (and hence any) $x_0 \in X$. (That this inequality holds for all x, if it should hold for some x_0, follows from the triangle inequality.) The set of all such probability measures is denoted by $\mathcal{P}_p(X)$. Additionally, the collection of all Borel probability measures, with or without finite moments, is denoted by $\mathcal{P}(X)$.

Essentially, a Borel probability measure μ has finite pth moment if the expected distance between any two points in its support is finite. This regularity assumption is needed to show that the Wasserstein distance (of order p) is an actual metric. If instead we work with the space of all Borel probability measures, the Wasserstein distance is only a generalized metric, meaning, roughly, that it takes values in $[0, \infty]$, as opposed to $[0, \infty)$.

Definition 2.5. Let $\mu \in \mathcal{P}_p(X)$, and let $\nu \in \mathcal{P}_p(Y)$. A **coupling** of μ and ν is a probability measure $\pi \in \mathcal{P}(X \times Y)$ with first and second marginals equal to μ and ν, respectively. That is, for all $A \in \Sigma_X$ and all $B \in \Sigma_Y$,

$$\pi(A \times Y) = \mu(A), \quad \text{and} \quad \pi(X \times B) = \nu(B).$$

The collection of all couplings between μ and ν is denoted by $\Pi(\mu, \nu)$.

A coupling can be seen as a relaxation of a transport map, which is a measurable map $T : X \to Y$ satisfying $T_\sharp \mu = \nu$ where $T_\sharp \mu$ is the **pushforward** of μ defined by

$$(T_\sharp \mu)(B) = \mu(T^{-1}(B))$$

for all $B \in \Sigma_Y$. The Monge optimal transport problem, first posed by Monge [40] in 1781, albeit not in these terms, asks for the minimum of

$$I[T] = \int_X c(x, T(x))\mu(dx)$$

over all transport maps T with $T_\sharp \mu = \nu$ where c is a measurable cost function on $X \times Y$. Kantorovich, in the 1940s [34, 35], instead considered minimizing

$$I[\pi] = \int_X c(x, y)\pi(dx, dy)$$

over all couplings π of μ and ν. This problem is much more tractable than Monge's. When c is a metric, we obtain the Wasserstein distance.

Definition 2.6. Let (X, d_X) be a complete separable metric space. The **Wasserstein metric of order p** $W_{p,d_X} : \mathcal{P}_p(X) \times \mathcal{P}_p(X) \to [0, \infty)$ is defined by

$$W_{p,d_X}(\mu, \nu) = \left(\inf_{\pi : \Pi(\mu,\nu)} \int_{X \times X} d_X^p(x_1, x_2)\pi(dx_1, dx_2) \right)^{1/p}.$$

(Note that we use the type-theoretic notation $\pi : \Pi(\mu, \nu)$, rather than $\pi \in \Pi(\mu, \nu)$, underneath the infinum for better readability.) We call the pair $(\mathcal{P}_p(X), W_{p,d_X})$ the **Wasserstein space of order p** associated with (X, d_X).

Informally, we say that the Wasserstein construction lifts any metric d_X to a metric W_{p,d_X} defined on (a subset of) the Borel probability measures over X. The metric measures the minimum amount of effort needed to reshape one probability mass into another. The coupling π tells us how to redistribute the mass of the initial measure μ into the target measure ν, and the (pth power of) the ground metric encodes the cost associated to any such rearrangement.

Later, we will show that any coupling π of μ and ν, optimal or otherwise, can itself be regarded as an element of a Wasserstein space. Therefore, in the same way that we measure the distance between measures μ and ν, we can also measure distances between couplings π_1 and π_2. In fact, this construction can be iterated: A coupling of couplings is again an element of a Wasserstein space, as is a coupling of a coupling of couplings, and so on. There are different metrics one can place on $X \times Y$ to realize this perspective. For our purposes, though, we always use the product metric.

Definition 2.7. Let (X, d_X) and (Y, d_Y) be Polish, and let $1 \leq p < \infty$. Then, the **product metric of order p**

$$d_{p,X\times Y} : (X \times Y) \times (X \times Y) \to \mathbb{R}_{\geq 0}$$

is defined by

$$d_{p,X\times Y}\big((x_1, y_1), (x_2, y_2)\big) = \big(d_X^p(x_1, x_2) + d_Y^p(y_1, y_2)\big)^{1/p}.$$

For simplicity, we denote this metric by d_{XY}. We will always choose p to match the order of the Wasserstein metric of interest.

The Wasserstein space $(\mathcal{P}_p(X \times Y), W_{p,d_{XY}})$ is simply the usual Wasserstein space where the ground metric is now d_{XY}. More abstractly, we can also consider spaces like $\mathcal{P}_p(\mathcal{P}_p(X) \times \mathcal{P}_p(Y))$ where the ground metric is $W_{p,d_X} \otimes W_{p,d_Y}$.

To prove that the Wasserstein distance is, in fact, a metric—that, in particular, the triangle inequality holds—one uses the following gluing lemma.

Lemma 2.8 (**Gluing Lemma**). *Let μ_j be a probability measure supported on a Polish space X_j for $j = 1, 2, 3$. Further, let $\pi_{12} \in \Pi(\mu_1, \mu_2)$ and let $\pi_{23} \in \Pi(\mu_2, \mu_3)$ be couplings. Then, there exists a probability measure $\pi \in \mathcal{P}(X_1 \times X_2 \times X_3)$ with marginals π_{12} on $X_1 \times X_2$ and π_{23} on $X_2 \times X_3$.*

Proof. See Lemma 7.6 in Villani [59]. □

We also need to know that the space of couplings is non-empty, and indeed it is, since, at the least, we can always take the product probability measure.

Definition 2.9. Let $\mu \in \mathcal{P}_p(X)$ and $\nu \in \mathcal{P}_p(Y)$. The **product measure**, or tensor product, of μ and ν is the joint probability measure $\mu \otimes \nu$ in $\mathcal{P}_p(X \times Y)$ defined on test functions $\phi \in C_b(X \times Y)$ by

$$\int_{X \times Y} \phi(x, y)(\mu \otimes \nu)(dx, dy) = \int_Y \left(\int_X \phi(x, y)\mu(dx) \right) \nu(dy).$$

Alternatively, we can declare

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B) \tag{1}$$

for $A \times B \in \Sigma_X \times \Sigma_Y$. Since $X \times Y$ is standard Borel, (1) determines $\mu \otimes \nu$ uniquely on all of $\Sigma_{X \times Y}$.

A further crucial property of the Wasserstein metric is that the infinimum is actually achieved; see Proposition 2.1 in Villani [59]. Hence, we can define the set $\Pi_o(\mu, \nu)$ of **optimal plans** to be all couplings $\pi \in \Pi(\mu, \nu)$ such that

$$\int_{X \times X} d_X^p(x_1, x_2)\pi(dx_1, dx_2) = W_{p,d_X}^p(\mu, \nu).$$

Additionally, we frequently appeal to the following result, which says that the Wasserstein space associated to a Polish metric space is again Polish.

Theorem 2.10. *Let (X, d_X) be a complete and separable metric space. Then, the Wasserstein space $(\mathcal{P}_p(X), W_{p,d_X})$ is also complete and separable.*

Proof. See Proposition 7.1.5 in Ambrosio et al. [2]. □

We now confirm that we can indeed view couplings as elements of a Wasserstein space. The proof is elementary, but the result is conceptually important.

Proposition 2.11. *Let $\mu \in \mathcal{P}_p(X)$ and let $\nu \in \mathcal{P}_p(Y)$. Let $\gamma \in \Pi(\mu, \nu) \subseteq \mathcal{P}(X \times Y)$ be any coupling of μ and ν. Then, $\gamma \in \mathcal{P}_p(X \times Y)$. That is, γ has finite pth moment with respect to the product metric d_{XY}. Additionally, $\Pi(\mu, \nu)$ and $\Pi_o(\mu, \nu)$ are both closed convex subsets of the Wasserstein space $\mathcal{P}_p(X \times Y)$.*

Proof. Fix $(x_0, y_0) \in X \times Y$, and let $\gamma \in \Pi(\mu, \nu)$ be a coupling of $\mu \in \mathcal{P}_p(X)$ and $\nu \in \mathcal{P}_p(Y)$. Then,

$$\begin{aligned}
&\int_{X\times Y} d_{XY}^p\big((x_0, y_0), (x_1, y_1)\big)\gamma(dx_1, dy_1) \\
&\quad = \int_{X\times Y} d_X^p(x_0, x_1) + d_Y^p(y_0, y_1)\gamma(dx_1, dy_1) \\
&\quad = \int_X d_X^p(x_0, x_1)\mu(dx_1) + \int_Y d_Y^p(y_0, y_1)\nu(dy_1) \\
&\quad < \infty.
\end{aligned}$$

Thus, $\gamma \in \mathcal{P}_p(X \times Y)$. Further, that $\Pi(\mu, \nu)$ and $\Pi_o(\mu, \nu)$ are both convex is essentially immediate. Also, if $\gamma_n \in \Pi(\mu, \nu)$ with $\gamma_n \to \gamma$ in the Wasserstein metric, then $\gamma \in \Pi(\mu, \nu)$, since the marginals are constant. If, in addition, each γ_n is optimal, then γ is too, since, in this case,

$$\int_X d_X^p(x_1, x_2)\gamma_n(dx_1, dx_2) = W_{p,d_X}^p(\mu, \nu)$$

is constant. □

When thinking of couplings as elements of a Wassersetein space, we must be careful to distinguish between the Wasserstein topology and the weak topology.

Definition 2.12. Let $\{\mu_n\}_{n=1}^\infty \subseteq \mathcal{P}(X)$ be a sequence of Borel probability measures. Then, $\{\mu_n\}_{n=1}^\infty$ **converges weakly** to $\mu \in \mathcal{P}(X)$ if for all $f \in C_b(X)$,

$$\int_X f(x)\mu_n(dx) \xrightarrow{n\to\infty} \int_X f(x)\mu(dx).$$

Further, we say that the **pth moments of $\{\mu_\mathbf{n}\}_{\mathbf{n=1}}^\infty$ converge** to the pth moment of μ if for some (and hence any) $x_0 \in X$,

$$\int_X d_X^p(x_0, x)\mu_n(dx) \xrightarrow{n\to\infty} \int_X d_X^p(x_0, x)\mu(dx) < \infty.$$

Notice that in this second case we assume the pth moment of μ is indeed finite.

Theorem 2.13. *Let $\{\mu_n\}_{n=1}^\infty \subseteq \mathcal{P}_p(X)$. Then, $W_{p,d_X}(\mu_n, \mu) \to 0$ if and only if $\{\mu_n\}_{n=1}^\infty$ converges weakly to μ and the pth moments of $\{\mu_n\}_{n=1}^\infty$ converge to the pth moment of μ.*

Proof. See Theorem 7.11 in Villani [59]. □

In a phrase, Theorem 2.13 says that the Wasserstein metric of order p metrizes p-finite weak convergence. The weak topology on $\mathcal{P}(X)$ is induced by weak convergence alone, whereas the Wasserstein topology on $\mathcal{P}_p(X)$ is equivalent to the topology induced by p-finite weak convergence. It is important to keep track of this distinction, since many theorems in the literature consider only the weak topology on $\mathcal{P}(X \times Y)$. To apply such theorems to the Wasserstein space $\mathcal{P}_p(X \times Y)$, it is enough to verify that we also have convergence of pth finite moments in any given scenario of interest.

To summarize, and to clarify notation, the situation is this: Any coupling $\pi \in \Pi(\mu,\nu)$ is automatically an element of $\mathcal{P}_p(X \times Y)$. $\Pi(\mu,\nu)$ is a closed convex subset of $\mathcal{P}(X \times Y)$ when equipped with the weak topology, and it is also a closed convex subset of the Wasserstein space $\mathcal{P}_p(X \times Y)$. In computations, we simply write $\Pi(\mu,\nu)$. We write $\Pi_p(\mu,\nu)$ when we want to emphasize that we are thinking of the space of couplings equipped with the Wasserstein subspace topology. Finally, $\Pi_o(\mu,\nu)$ denotes the set of optimal plans, and we write $\Pi_p^o(\mu,\nu)$ if we again wish to emphasize that we are working in the Wasserstein subspace topology.

Finally, we give the statement of Kantorovich duality, which is essential for many of our proofs, and we also state two measurable selection theorems that we will need.

Theorem 2.14 (**Kantorovich Duality**). *Let X and Y be Polish spaces, and let $c : X \times Y \to [0,\infty]$ be a lower semi-continuous cost function. Then, for any $\mu \in \mathcal{P}(X)$ and any $\nu \in \mathcal{P}(Y)$,*

$$\inf_{\pi:\Pi(\mu,\nu)} \int_{X\times Y} c(x,y)\pi(dx,dy) = \sup_{(\phi,\nu):\Phi_c(\mu,\nu)} \int_X \phi(x)\mu(dx) + \int_Y \psi(y)\nu(dy),$$

where $\Phi_c(\mu,\nu)$ is defined as all functions $(\phi,\psi) \in L^1(X,\mu) \times L^1(Y,\nu)$ satisfying

$$\phi(x) + \psi(y) \leq c(x,y)$$

for μ-a.e. $x \in X$ and ν-a.e. $y \in Y$. In fact, the above holds when restricting Φ_c to test functions $(\phi,\psi) \in C_b(X) \times C_b(Y)$.

Proof. See Theorem 1.3 in Villani [59]. □

In particular, we have

$$W^p_{p,d_X}(\mu,\nu) = \sup_{(\phi,\nu):\Phi_c(\mu,\nu)} \int_X \phi(x)\mu(dx) + \int_X \psi(x)\nu(dx)$$

for $c = d_X^p$. When the context is clear, we often write Φ_c or Φ in place of $\Phi_c(\mu,\nu)$.

Theorem 2.15 (**Measurable Selection Theorem**). *Let X be Polish, and let (Ω,Σ_Ω) be a measurable space. Let $\psi : \Omega \to Cl(X)$ take values in the set $Cl(X)$ of non-empty, closed subsets of X. Assume that ψ is Σ_Ω-weakly measurable, meaning that for every open $U \subseteq X$,*

$$\{\omega : \psi(\omega) \cap U \neq \emptyset\} \in \Sigma_\Omega.$$

Then, ψ has a Σ_Ω-Σ_X-measurable selection, that is, a measurable function $\phi : \Omega \to X$ with $\phi(x) \in \psi(x)$.

Theorem 2.16 (**Measurable Selection of Optimal Plans**). *Let X and Y be Polish spaces, and let $c : X \times Y \to \mathbb{R}$ be a continuous cost function with $\inf c > -\infty$. Let Ω be a measurable space, and let $\omega \mapsto (\mu_\omega,\nu_\omega)$ be a measurable function $\Omega \to \mathcal{P}(X) \times \mathcal{P}(Y)$. Then, there is a measurable choice $\omega \mapsto \pi_\omega$ such that π_ω is an optimal transference plan between μ_ω and ν_ω for each $\omega \in \Omega$.*

Proof. See Corollary 5.22 of Villani [60]. □

2.2. **Markov Kernels.** Next we introduce the necessary material on Markov kernels.

Definition 2.17. Let (X, Σ_X) and (Y, Σ_Y) be measurable spaces. A **Markov kernel** from (X, Σ_X) to (Y, Σ_Y) is a map $F : \Sigma_Y \times X \to [0, 1]$ with the following properties:

(1) $F(-|x) : \Sigma_Y \to [0, 1]$ is a probability measure for any fixed $x \in X$.
(2) The map $x \mapsto F(E|x)$ is Σ_X-measurable for any fixed $E \in \Sigma_Y$.

Note that the conditional notation $F(-|x)$ is simply alternative notation for $F(-, x)$.

A Markov kernel can equivalently be viewed as a measurable map $F : X \to \mathcal{P}(Y)$, where $\mathcal{P}(Y)$ is equipped with the initial σ-algebra with respect to all evaluation maps $\mathrm{ev}_E : \mathcal{P}(Y) \to [0, 1]$ for $E \in \Sigma_Y$; see, e.g., Lemma 3.1 in [33]. In the main result of this subsection, Theorem 2.25, we show that we can similarly view a Markov kernel F with values in $\mathcal{P}_p(Y)$ as a measurable map $F : X \to \mathcal{P}_p(Y)$, where $\mathcal{P}_p(Y)$ is equipped with the Borel σ-algebra induced by the Wasserstein metric. In the meantime, we also use the notation $F : X \to \mathcal{P}_p(Y)$ to denote an ordinary Markov kernel, in the sense of Definition 2.17, with the added condition that $F(-|x) \in \mathcal{P}_p(Y)$ for all $x \in X$.

As with the space of measures, we must impose regularity assumptions on Markov kernels to ensure that we have an actual metric.

Definition 2.18. Let (X, d_X, μ) be a metric measure space, and let (Y, d_Y) be Polish. A Markov kernel $F : X \to \mathcal{P}_p(Y)$ has **finite pth moment** if

$$\int_X \left(\int_Y d_Y^p(y_0, y) F(dy|x) \right) \mu(dx) < \infty$$

for some (and hence any) $y_0 \in Y$. Let $\mathcal{K}_\mu^p(X, Y)$ denote the collection of all such Markov kernels.

We will equip Markov kernels (having finite pth moment) with a Wasserstein-type metric first defined by Patterson in [45].

Definition 2.19. A **coupling** of Markov kernels $F : X \to \mathcal{P}(Y)$ and $G : X \to \mathcal{P}(Z)$ is a Markov kernel $\Gamma : X \to Y \times Z$ with marginals given by F and G. Equivalently, Γ_x is a coupling of F_x and G_x. The set of all couplings is denoted by $\Pi(F, G)$.

Definition 2.20 (Definition 5.2 in [45])**.** Let (X, d_X, μ) be a metric measure space, let (Y, d_Y) be Polish, and let $1 \leq p < \infty$. The **Markov-Wasserstein metric of order p** $MW_p : \mathcal{K}_\mu^p(X, Y)^2 \to [0, \infty)$ is defined by

$$MW_p(F, G) = \left(\inf_{\Gamma : \Pi(F,G)} \int_{X \times Y^2} d_Y^p(y_1, y_2) \Gamma(dy_1, dy_2|x) \mu(dx) \right)^{1/p}.$$

When X is a point, the Markov-Wasserstein distance reduces to the ordinary Wasserstein distance, and it reduces to L^p distance when the Markov kernels are deterministic. In practice, we will only ever use an equivalent form of the Markov-Wasserstein metric derived in Lemma 2.22, which shows that the Markov-Wasserstein metric is actually an instance of the L^p metric for metric-space valued functions.

Definition 2.21. Let (X, d_X, μ) be a metric measure space, and let (Y, d_Y) be Polish. The **$\mathbf{L^p}$ space** $L^p_\mu(X, Y)$ is the set of μ-a.e. equal equivalence classes of measurable functions $f : X \to Y$ satisfying

$$\int_X d_Y^p\big(y_0, f(x)\big)\mu(dx) < \infty$$

for some (and hence any) $y_0 \in Y$. A function satisfying this condition is said to be **$\mathbf{L^p}$ integrable**. $L^p_\mu(X, Y)$ is a metric space when equipped with the **$\mathbf{L^p}$ metric**:

$$d_{L^p}(f, g) = \left(\int_X d_Y^p\big(f(x), g(x)\big)\mu(dx)\right)^{1/p}$$

for $1 \le p < \infty$, and for $p = \infty$,

$$d_{L^\infty}(f, g) = \operatorname*{ess\,sup}_{x\in X} d_Y\big(f(x), g(x)\big).$$

Lemma 2.22 (Proposition 5.5 in [45])**.** *Let $F, G \in \mathcal{K}^p_\mu(X, Y)$. Then,*

$$MW_p(F, G) = \left(\int_X W^p_{p,d_Y}(F_x, G_x)\mu(dx)\right)^{1/p}.$$

Proof. By Corollary 2.23, there exists $\sigma : X \to \mathcal{P}(Y \times Y)$ such that $\sigma_x \in \Pi(F_x, G_x)$, and

$$W^p_{p,d_Y}(F_x, G_x) = \int_{Y^2} d_Y^p(y_1, y_2)\sigma_x(dy_1, dy_2).$$

Hence,

$$\begin{aligned}
MW_p^p(F, G) &= \inf_{\pi:\Pi(F,G)} \left(\int_{X\times Y^2} d_Y^p(y_1, y_2)\pi_x(dy_1, dy_2)\right)\mu(dx) \\
&= \int_X \left(\inf_{\gamma:\Pi(F_x,G_x)} \int_{Y^2} d_Y^p(y_1, y_2)\gamma(dy_1, dy_2)\right)\mu(dx) \\
&= \int_X \left(\int_{Y^2} d_Y^p(y_1, y_2)\sigma_x(dy_1, dy_2)\right)\mu(dx) \\
&= \int_X W^p_{p,d_Y}(F_x, G_x)\mu(dx).
\end{aligned}$$

□

Note also that we still have measurable selections of optimal couplings in the Markov-kernel setting.

Theorem 2.23 (**Existence of Optimal Couplings**)**.** *Let $F, G \in \mathcal{K}_\mu(X, Y)$, and let $c : Y \times Y \to \mathbb{R}_+$ be a nonnegative, lower semicontinuous cost function. Then, there exists a Markov kernel $\pi : X \times X \to \mathcal{P}(Y \times Y)$ such that*

(1) π_{x_1,x_2} is a coupling of F_{x_1} and G_{x_2}.
(2) Additionally,

$$\int_{Y^2} c(y_1, y_2)\pi_{x_1,x_2}(dy_1, dy_2) = \inf_{\gamma:\Pi(F_{x_1},G_{x_2})} \int_{Y^2} c(y_1, y_2)\gamma(dy_1, dy_2).$$

Now, our goal, recall, is to show that a Markov kernel F with values in $\mathcal{P}_p(Y)$ can equivalently be regarded as a Borel map into the Wasserstein space. If we could show that the σ-algebra Σ_W induced by the Wasserstein topology is equivalent to the initial σ-algebra Σ_{ev}, then the proof of equivalence in the purely measure-theoretic setting would apply immediately. Lemma 2.24 below implies that $\Sigma_{\text{ev}} \subseteq \Sigma_W$. However, proving the other inclusion in a hands-on manner faces certain additional challenges related to the fact that the Wasserstein metric only metrizes p-finite weak convergence, as opposed to weak convergence, more generally. Lemma 12.4.7 in Ambrosio et al. [2], together with Lemma 2.24 below, allows us to achieve the desired result nonetheless.

Lemma 2.24. *Let (X, d_X) be a Polish metric space, let $(\mathcal{P}_p(X), W_{p,d_X})$ be its corresponding Wasserstein space, and let Σ_W be the Borel σ-algebra on $\mathcal{P}_p(X)$ induced by the Wasserstein metric. Then, for any $A \in \Sigma_X$, the evaluation map $ev_A : \mathcal{P}_p(X) \to [0,1]$ given by $ev_A(\mu) = \mu(A)$ is Σ_W-measurable.*

Proof. Let $\mathcal{A} \subseteq \Sigma_X$ be all Borel sets $A \in \Sigma_X$ such that $\text{ev}_A : \mathcal{P}_p(X) \to [0,1]$ is Σ_W-measurable. Note that $\mathcal{A}$ is non-empty since, for example, $X \in \mathcal{A}$. We will apply the Monotone Class Theorem to show that $\mathcal{A} = \Sigma_X$. Let $\mathcal{C}$ be all Borel sets $C \in \Sigma_X$ such that the indicator function $\mathbf{1}_C$ can be written as the pointwise limit of functions $\{f_n\}_{n=1}^{\infty}$ with $f_n \in C_b(X)$ and $0 \leq f_n \leq 1$. In particular, $\mathcal{C}$ includes all open sets. We'll first show that $\mathcal{C} \subseteq \mathcal{A}$ and that $\mathcal{C}$ is an algebra.

In fact, for any $C \in \mathcal{C}$, ev_C is continuous. To see this, let $B \subseteq [0,1]$ be open. We can write $B = \bigcup_{k=1}^{\infty} B_{r_k}(z_k)$ for $r_k > 0$ and $z_k \in (0,1)$. Set $E \coloneqq \text{ev}_C^{-1}(B)$, and fix some $\mu_0 \in E$. We will construct a Wasserstein ball

$$W_\delta(\mu_0) = \{\mu \in \mathcal{P}_p(X) : W_{p,d_X}(\mu, \mu_0) < \delta\}$$

with $W_\delta(\mu_0) \subseteq E$. Take k such that $\mu_0(C) \in B_{r_k}(z_k)$, and take $s > 0$ such that $B_s\big(\mu_0(C)\big) \subseteq B_{r_k}(z_k)$. Set $F \coloneqq \text{ev}_C^{-1}\big(B_s(\mu_0(C))\big)$. Then, $F \subseteq E$. Since the Wasserstein distance metrizes p-finite weak convergence, we have that for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $f \in C_b(X)$,

$$\left| \int_X f(x)\mu(dx) - \int_X f(x)\mu_0(dx) \right| < \varepsilon$$

whenever $W_{p,d_X}(\mu, \mu_0) < \delta$. Further, by the Dominated Convergence Theorem,

$$\text{ev}_C(\mu) = \int_X \mathbf{1}_C(x)\mu(dx) = \lim_{n\to\infty} \int_X f_n(x)\mu(dx).$$

Thus, given $\varepsilon \coloneqq s/3 > 0$, there exists $\delta > 0$ such that for any $\mu \in W_\delta(\mu_0)$, we can take N sufficiently large so that

$$\begin{aligned} |\mu(C) - \mu_0(C)| &= \left| \int_X \mathbf{1}_C(x)\mu(dx) - \int_X \mathbf{1}_C(x)\mu_0(dx) \right| \\ &\leq \left| \int_X \mathbf{1}_C(x) - f_N(x)\mu(dx) \right| + \left| \int_X f_N(x)\mu(dx) - \int_X f_N(x)\mu_0(dx) \right| \end{aligned}$$

$$
\begin{aligned}
&\qquad\qquad + \left| \int_X f_N(x) - \mathbf{1}_C(x)\mu_0(dx) \right| \\
&< \frac{s}{3} + \frac{s}{3} + \frac{s}{3} \\
&= s.
\end{aligned}
$$

This gives us $W_\delta(\mu_0) \subseteq F \subseteq E$. As such, E is open, which shows that ev_C is continuous and hence measurable. Thus, $\mathcal{C} \subseteq \mathcal{A}$.

$\mathcal{C}$ is also an algebra. First, X is in $\mathcal{C}$, since the indicator function $\mathbf{1}_X$ is simply the constant function $f(x) = 1$. Further, if $C_1 \in \mathcal{C}$ with $\{f_n\}_{n=1}^\infty$ converging pointwise to $\mathbf{1}_C$, then $\{1 - f_n\}_{n=1}^\infty$ converges to $\mathbf{1}_{C^c}$, meaning $\mathcal{C}$ is closed under complements. Finally, if C_2 is any other element of $\mathcal{C}$, then $C_1 \cap C_2 \in \mathcal{C}$, since $\left\{(f_n)_{|C_1 \cap C_2}\right\}_{n=1}^\infty$ converges to $\mathbf{1}_{C_1 \cap C_2}$ with the necessary properties.

All that remains is to show that $\mathcal{A}$ is a monotone class. That is, for any increasing sequence of sets $A_n \in \mathcal{A}$, we need $U := \bigcup_{n=1}^\infty A_n \in \mathcal{A}$. Additionally, we need $I := \bigcap_{n=1}^\infty A_n \in \mathcal{A}$ for any decreasing sequence $A_n \in \mathcal{A}$. Indeed, by continuity from below,

$$
\mathrm{ev}_U(\mu) = \mu\left(\bigcup_{n=1}^\infty A_n\right) = \lim_{n\to\infty} \mu(A_n) = \lim_{n\to\infty} \mathrm{ev}_{A_n}(\mu),
$$

and since the pointwise limit of Borel measurable functions is Borel measurable, $U \in \mathcal{A}$. Similarly, closure under intersections of decreasing sequences follows from continuity from above.

We now have that $\mathcal{A}$ is a monotone class, and it contains an algebra $\mathcal{C}$. By Theorem 1.9.3 in [9], the monotone class $m(\mathcal{C})$ generated by $\mathcal{C}$ coincides with the σ-algebra $\sigma(\mathcal{C})$ generated by $\mathcal{C}$. Then, since the class of open sets $\mathcal{B}$ is contained in $\mathcal{C}$, we have

$$
\Sigma_X = \sigma(\mathcal{B}) \subseteq \sigma(\mathcal{C}) = m(\mathcal{C}) \subseteq \mathcal{A}.
$$

Thus, $\Sigma_X = \mathcal{A}$.

□

Theorem 2.25. *Let (X, d_X, μ) be a metric measure space, let (Y, d_Y) be Polish, and let $1 \le p < \infty$. Then, the metric spaces $(\mathcal{K}_\mu^p(X, Y), MW_p)$ and $(L_\mu^p(X, \mathcal{P}_p(Y)), d_{L^p})$ are isometrically isomorphic.*

Proof. Lemma 12.4.7 in [2] shows that $F : X \to \mathcal{P}_p(Y)$ is a Borel map whenever $F : \Sigma_Y \times X \to [0, 1]$ is a Markov kernel. Conversely, if $F : X \to \mathcal{P}_p(Y)$ is Borel measurable, then it trivially satisfies the first condition of being a Markov kernel. Further, by Lemma 2.24, $\mathrm{ev}_B : \mathcal{P}_p(Y) \to [0, 1]$ is measurable for any $B \in \Sigma_Y$, and since the composition of measurable maps is measurable, $\mathrm{ev}_B \circ F$ is measurable, as needed.

Additionally, we have that a Markov kernel F has finite pth moment if and only if it is L^p integrable when viewed as a Borel map. Indeed, let $y_0 \in Y$. Then, since there is a unique coupling between any measure and a single Dirac mass, we have

$$
\int_X W_{p,d_Y}^p\big(\delta_{y_0}, F(x)\big)\mu(dx) = \int_X \left(\inf_{\pi:\Pi(\delta_{y_0}, F_x)} \int_{Y^2} d_Y^p(y_1, y_2)\pi(dy_1, dy_2) \right) \mu(dx)
$$

$$= \int_X \left(\int_{Y^2} d_Y^p(y_1, y_2)(\delta_{y_0} \otimes F_x)(dy_1, dy_2) \right) \mu(dx)$$
$$= \int_X \left(\int_Y d_Y^p(y_0, y) F_x(dy) \right) \mu(dx).$$

The process of passing from an ordinary Markov kernel to a Borel map, and vice versa, is unique up to sets of measure zero. Thus, the sets $\mathcal{K}_\mu^p(X, Y)$ and $L_\mu^p(X, \mathcal{P}_p(Y))$ are in one-to-one correspondence. By Lemma 2.22, they are isometrically isomorphic as metric spaces when equipped with the Markov-Wasserstein metric and the L^p metric, respectively.

□

Throughout the rest of the work, we will think of Markov kernels as Borel maps into a Wasserstein space. We will denote the metric by MW_p, but we will always compute the metric in its L^p form as an integral of a Wasserstein metric.

3. Integration and Disintegration

In this section, we develop a Lebesgue-like theory of integration for Markov kernels and prove a modified version of the Disintegration Theorem. In essence, the ordinary Disintegration Theorem allows us to decompose a measure on some space Y into an X-indexed collection of measures on Y (each supported on a set of measure zero), where Y and X are related via some projection map $p : Y \to X$. The original measure can then be recovered by gluing the parameterized measures together via integration. Or, at least, this is the intuition for the result.

In actuality, if $\nu \in \mathcal{P}_p(Y)$ is the measure of interest, and if $\{\nu_x\}_{x \in X} \subseteq \mathcal{P}_p(Y)$ is the parameterized collection of measures, then, in fact, the way we reconstruct ν is by defining a functional on $C_b(Y)$ via an iterated integral. That is, for any $\phi \in C_b(Y)$, we can show that

$$\int_Y \phi(y)\nu(dy) = \int_X \left(\int_Y \phi(y)\nu_x(dy) \right) (p_\sharp \nu)(dx),$$

where $p_\sharp \nu(A) = \nu(p^{-1}(A))$ for $A \in \Sigma_X$ is the pushforward measure on X.

Here, we show that the intuitive picture first presented can be made rigorous. The parameterized collection of measures $\{\nu_x\}_{x \in X}$ is equivalent to a Markov kernel $F : X \to \mathcal{P}_p(Y)$, and we have shown that Markov kernels, in turn, are equivalent to measure-valued Borel maps. Hence, we develop a theory of a Lebesgue-like integral, which we call the Markov-Lebesgue integral, for these measure-valued maps, and we show that the measure defined by the Markov-Lebesgue integral is equivalent to the measure defined by the iterated integral. As such, we can then show that the Disintegration Theorem can be interpreted as a literal equality of (Markov-Lebesgue) integrals. Besides being of conceptual importance, the approximation machinery developed for this result is what allows us to give a simplified proof of a generalized Bayes's Law in Theorem 5.4.

3.1. **Integration.** To define the Markov-Lebesgue integral, naturally we start with a definition of simple Markov kernels. Since we only ever integrate against probability measures, we elect to use countable, rather than finite, simple maps as our basic building blocks. Admitting countable sums allows us to prove uniform, rather than pointwise, convergence of simple Markov kernels to general Markov kernels, and this greatly simplifies the proofs. Equipped with convergence results and with a definition of the Markov-Lebesgue integral for simple Markov kernels, we then define the general Markov-Lebesgue integral as the limit of a certain Cauchy sequence.

Definition 3.1. A Markov kernel $F : X \to \mathcal{P}_p(Y)$ is called **countably simple** if it is of the form

$$F = \sum_{j=1}^{\infty} \nu_j \mathbf{1}_{E_j},$$

where $\{E_j\}_{j=1}^{\infty} \subseteq \Sigma_X$ is a pairwise disjoint collection of measurable sets with $\bigcup_{j=1}^{\infty} E_j = X$, and where each $\nu_j \in \mathcal{P}_p(Y)$. (Note that countably simple Markov kernels include finite simple Markov kernels, since we can set $E_j = \emptyset$ for all j greater than some N.)

The definition of the integral for countably simple Markov kernels is immediate.

Definition 3.2. Let $\mu \in \mathcal{P}_p(X)$ be a probability measure, and let $F = \sum_{j=1}^{\infty} \nu_j \mathbf{1}_{E_j}$ be a countably simple L^p_μ Markov kernel; (that is, F is a countably simple Markov kernel that is also L^p integrable with respect to μ). Then, the **Markov-Lebesgue integral** of F against μ is the probability measure in $\mathcal{P}_p(Y)$ defined by

$$\int_X F(x)\mu(dx) := \sum_{j=1}^{\infty} \mu(E_j)\nu_j.$$

We will also denote the integral by $\int_X F d\mu$ or, in some cases, simply by $I_\mu(F)$. Note also that when the measure μ is clear from context, we call F an L^p Markov kernel and drop μ from the notation.

In the next proposition, we verify that $\int_X F d\mu$ is indeed an element of $\mathcal{P}_p(Y)$.

Proposition 3.3. *Let $\mu \in \mathcal{P}_p(X)$ be a probability measure, and let $F = \sum_{j=1}^{\infty} \nu_j \mathbf{1}_{E_j}$ be a countably simple L^p Markov kernel. Then, the Markov-Lebesgue integral $\int_X F d\mu$ of F is an element of $\mathcal{P}_p(Y)$. Additionally, for any $\phi \in C_b(Y)$,*

$$\int_Y \phi(y) \left(\int_X F d\mu \right)(dy) = \int_X \left(\int_Y \phi(y) F_x(dy) \right) \mu(dx). \tag{2}$$

That is, the probability measure defined by the Markov-Lebesgue integral is equal to the measure defined by the iterated integral.

Proof. First, note that

$$\left(\int_X F d\mu \right)(Y) = \sum_{j=1}^{\infty} \mu(E_j)\nu_j(Y) = \sum_{j=1}^{\infty} \mu(E_j) = 1,$$

and similarly, $\left(\int_X F d\mu\right)(\emptyset) = 0$. Additionally, if $\{B_i\}_{i=1}^{\infty}$ is a countable collection of disjoint measurable sets in Σ_Y, then, by absolute convergence,

$$\begin{aligned}\left(\int_X F d\mu\right)\left(\bigcup_{i=1}^{\infty} B_i\right) &= \sum_{j=1}^{\infty} \mu(E_j)\nu_j\left(\bigcup_{i=1}^{\infty} B_i\right)\\ &= \sum_{j=1}^{\infty} \mu(E_j) \sum_{i=1}^{\infty} \nu_j(B_i)\\ &= \sum_{i=1}^{\infty}\left(\sum_{j=1}^{\infty} \mu(E_j)\nu_j(B_i)\right)\\ &= \sum_{i=1}^{\infty}\left(\int_X F d\mu\right)(B_i).\end{aligned}$$

Thus, $\int_X F d\mu$ is a probability measure.

Next, let $\nu \in \mathcal{P}_p(Y)$ denote the probability measure determined by the following functional: For $\phi \in C_b(Y)$,

$$L(\phi) = \int_X \left(\int_Y \phi(y) F_x(dy)\right) \mu(dx).$$

Let $B \subseteq Y$ be open. Then, $\mathbf{1}_B$ can be written as the pointwise limit of functions $\phi_n \in C_b(Y)$ with $0 \leq \phi_n \leq 1$, and

$$\begin{aligned}\nu(B) &= \int_Y \mathbf{1}_B(y)\nu(dy)\\ &= \int_Y \lim_{n\to\infty} \phi_n(y)\nu(dy)\\ &= \lim_{n\to\infty} \int_Y \phi_n(y)\nu(dy)\\ &= \lim_{n\to\infty} L(\phi_n)\\ &= \int_X \left(\int_Y \mathbf{1}_B(y) F_x(dy)\right) \mu(dx).\end{aligned}$$

Hence,

$$\begin{aligned}\left(\int_X F d\mu\right)(B) &= \sum_{j=1}^{\infty} \mu(E_j)\nu_j(B)\\ &= \sum_{j=1}^{\infty} \mu(E_j) \int_Y \mathbf{1}_B(y)\nu_j(dy)\\ &= \sum_{j=1}^{\infty} \int_{E_j} \left(\int_Y \mathbf{1}_B(y)\nu_j(dy)\right) \mu(dx)\end{aligned}$$

$$= \int_X \left(\int_Y \mathbf{1}_B(y) F_x(dy) \right) \mu(dx)$$
$$= \nu(B).$$

By Lemma 7.1.2 in Bogachev [9], two finite Borel measures are equal if they coincide on all open sets. Thus, $\int_X F d\mu = \nu$. That is, the probability measure defined by the Markov-Lebesgue integral of a countably simply Markov kernel is equal to the probability measure determined by the iterated integral. We use this fact to verify that $\int_X F d\mu$ has finite pth moment.

Fix some $y_0 \in Y$. Then, using the equality established in the proof of Theorem 2.25, we have

$$\begin{aligned} \int_Y d_Y^p(y_0, y) \left(\int_X F d\mu \right)(dy) &= \int_X \left(\int_Y d_Y^p(y_0, y) F_x(dy) \right) \mu(dx) \\ &= \int_X W_{p,d_Y}^p \big(\delta_{y_0}, F(x)\big) \mu(dx) \\ &< \infty, \end{aligned}$$

as needed. □

Since we admit countable sums for simple Markov kernels, it is indeed necessary to impose some kind of additional integrability or regularity condition on F in order for $\int_X F d\mu$ to have finite pth moment. That is, we have

$$\int_X \left(\int_Y d_y^p(y_0, y) F_x(dy) \right) \mu(dx) = \sum_{j=1}^{\infty} \mu(E_j) \int_Y d_Y^p(y_0, y) \nu_j(dy).$$

Hence, taking a finite sum would suffice to show that the Markov-Lebesgue integral has finite pth moment. For countable sums, we have shown that L^p integrability is sufficient for the series on the right to converge, though there might also be weaker conditions worth considering.

Typically, the definition of the ordinary Lebesgue integral involves taking a supremum over simple functions. For our purposes, it is more appropriate to follow the approach taken by Bogachev in [9]. In definition 2.3.1, Bogachev calls a sequence $\{f_n\}_{n=1}^{\infty} \subseteq L^1(\mu)$ of real-valued functions **fundamental in the mean** if for all $\varepsilon > 0$, there is N such that for all $n, m \geq N$,

$$\int_X |f_n(x) - f_m(x)| \mu(dx) < \varepsilon.$$

This notion is what leads to an alternative definition of the Lebesgue integral that bypasses the need to take a supremum over simple functions. The obvious analog of this condition for the case of Markov kernels is stated below.

Definition 3.4. Let $\{F_n\}_{n=1}^{\infty}$ be a sequence of Markov kernels in $L_\mu^p(X, \mathcal{P}_p(Y))$. Then, $\{F_n\}_{n=1}^{\infty}$ is called **weakly Cauchy in the $\mathbf{L^p}$ metric**, or **weakly Cauchy** for short, if for all $\varepsilon > 0$, there exists N such that for all $n, m \geq N$, we have

$$\int_X W_{p,d_Y}^p \big(F_n(x), F_m(x)\big) \mu(dx) < \varepsilon.$$

Proposition 3.5. *For any Markov kernels $F, G \in L^p_\mu(X, \mathcal{P}_p(Y))$, we have*

$$W^p_{p,d_Y}\left(\int_X F(x)\mu(dx), \int_X G(x)\mu(dx)\right) \leq \int_X W_{p,d_Y}\big(F(x), G(x)\big)\mu(dx),$$

where the measures on the left-hand side are defined via the iterated integral.

Proof. The proof is the same as for Theorem 4.8 in Villani [60]. □

Proposition 3.3 shows that the Markov-Lebesgue integral coincides with the iterated integral for countably simply L^p Markov kernels. Thus, given a sequence $\{F_n\}_{n=1}^\infty$ of countably simple Markov kernels in $L^p_\mu(X, \mathcal{P}_p(Y))$, we have

$$W^p_{p,d_Y}\left(\int_X F_n(x)\mu(dx), \int_X F_m(x)\mu(dx)\right) \leq \int_X W^p_{p,d_Y}\big(F_n(x), F_m(x)\big)\mu(dx) \tag{3}$$

where now the measures on the left are defined via the Markov-Lebesgue integral. We will use this bound to define the general Markov-Lebesgue integral as the limit of a Cauchy sequence. First, though, we prove our fundamental approximation result in Theorem 3.9. Nested partitions are the key tool used in the proof, and in Lemma 3.7, we give an explicit construction for nested partitions with resolution decreasing to zero. Nested partitions will also be a key ingredient in our main convergence result for the expected Bayesian update, namely Theorem 5.10.

Definition 3.6. Let (X, d) be a Polish metric space with Borel σ-algebra Σ_X. A (countable) **partition** of $U \subseteq X$ is a collection $\mathcal{E} = \{E_n\}_{n=1}^\infty$ of pairwise disjoint Borel sets such that $\bigcup_{j=1}^\infty E_n = U$. The **diameter** of a measurable set $E \in \Sigma_X$ is $\mathrm{Diam}(E) := \sup_{x,y\in E} d_X(x, y)$, and the **resolution** of a partition $\mathcal{E}$ is given by $\mathrm{Res}(\mathcal{E}) := \sup_{E\in\mathcal{E}} \mathrm{Diam}(E)$. A partition $\mathcal{E}'$ is a **refinement** of $\mathcal{E}$ if for each $E' \in \mathcal{E}'$, there is (a unique) $E \in \mathcal{E}$ with $E' \subseteq E$. Finally, a **nested partition** (of U) is a sequence of partitions $\mathcal{E}_n$ such that for each n, $\mathcal{E}_{n+1}$ is a refinement of $\mathcal{E}_n$.

Note that a partition $\mathcal{E}'$ is a refinement of $\mathcal{E}$ if and only if for any $E \in \mathcal{E}$, there are elements $E'_1, E'_2, \ldots$ of $\mathcal{E}'$ such that $E = E'_1 \cup E'_2 \cup \cdots$. We use this characterization of a refinement of $\mathcal{E}$ in the following lemma.

Lemma 3.7 (**Nested Partitions**)**.** *Let (X, d) be a Polish metric space. Then, there exists a nested partition $\mathcal{E} = \{\mathcal{E}_n\}_{n=1}^\infty$ of X such that $Res(\mathcal{E}_n) \to 0$ as $n \to \infty$.*

Proof. Since (X, d) is separable, there exists a countable dense subset $D = \{x_j\}_{j\in I_1}$ of X with $I_1 \subseteq \mathbb{Z}_+$ such that for all $\varepsilon > 0$ and all $x \in X$, there exists $x_j \in D$ with $d(x, x_j) < \varepsilon$. Thus,

$$X = \bigcup_{j\in I_1} B\big(1, x_j\big),$$

where $B(1, x_j) =: U_j$ is the open ball of radius 1 centered at $x_j \in D$. Denote this open cover by $\mathcal{U}_1 := \{U_j\}_{j\in I_1}$, and note that, similarly, we can take an open cover consisting of open balls of any fixed radius centered at the elements of D.

When equipped with the subspace topology and the restricted metric, each U_j is itself a separable metric space. Thus, we can take $x_{jk} \in D$ and $U_{jk} := B\big(1/2, x_{jk}\big) \cap$

U_j such that $U_j = U_{j1} \cup U_{j2} \cup \cdots$. Thus, we have another countable open cover

$$\mathcal{U}_2 \coloneqq \bigcup_{(j,k)\in I_2} \{U_{jk}\}$$

of X, where $I_2 \subseteq \mathbb{Z}_+^2$. More generally, let $2 \leq n < \infty$, and let $U_{j_1,\ldots,j_n}$ denote a set $U \in \mathcal{U}_n$ satisfying

$$U_{j_1\ldots,j_n} \subseteq U_{j_1,\ldots,j_{n-1}} \subseteq \cdots \subseteq U_{j_1}.$$

For each $(j_1,\ldots,j_n) \in I_n \subseteq \mathbb{Z}_+^n$, take

$$U_{j_1,\ldots,j_n,k} \coloneqq B\left(\frac{1}{n+1}, x_{j_1,\ldots,j_n,k}\right) \cap U_{j_1,\ldots,j_n}$$

such that

$$U_{j_1,\ldots,j_n} = U_{j_1,\ldots,j_n,1} \cup U_{j_1,\ldots,j_n,2} \cup \cdots$$

where each $x_{j_1,\ldots,j_n,k}$ is some element of D. From this, we construct a new index set $I_{n+1} \subseteq \mathbb{Z}_+^{n+1}$, and we define

$$\mathcal{U}_{n+1} \coloneqq \bigcup_{(j_1,\ldots,j_n,j_{n+1})\in I_{n+1}} \{U_{j_1,\ldots,j_n,j_{n+1}}\}.$$

Once again, by construction, $\mathcal{U}_{n+1}$ is a countable open cover of X. (That each $U \in \mathcal{U}_{n+1}$ is open in X, as opposed to merely some subspace topology, follows from induction, starting from the fact that each U_j is open.)

We now use $\mathcal{U}_n$ to construct a countable partition $\mathcal{E}_n \subseteq \Sigma_X$ of X. Set

$$\begin{aligned} E_{j_1,\ldots,j_n} \coloneqq U_{j_1,\ldots,j_n} \setminus \Big(& U_{j_1,\ldots,(j_n-1)} \cup U_{j_1,\ldots,(j_n-2)} \cup \cdots \cup U_{j_1,\ldots,j_{(n-1)},1} \cup \\ & U_{j_1,\ldots,(j_{(n-1)}-1)} \cup U_{j_1,\ldots,(j_{(n-1)}-2)} \cup \cdots \cup U_{j_1,\ldots,j_{(n-2)},1} \cup \\ & \vdots \\ & U_{j_1,(j_2-1)} \cup U_{j_1,(j_2-2)} \cup \cdots \cup U_{j_1,1} \cup \\ & U_{(j_1-1)} \cup U_{(j_1-2)} \cup \cdots \cup U_1 \Big). \end{aligned}$$

For example,

$$E_{343} = U_{343} \setminus (U_{342} \cup U_{341} \cup U_{33} \cup U_{32} \cup U_{31} \cup U_2 \cup U_1).$$

This generalizes the standard process for making an open cover disjoint. It is clear that $\mathcal{E}_1$ is a pairwise disjoint partition of X. To see that $\{\mathcal{E}_n\}_{n=1}^\infty$ is a nested partition, let $J = (j_1,\ldots,j_n)$. Then, we can write $U_J = \bigcup_{k\in I_J} U_{J,k} = \bigcup_{k\in I_J} U_{j_1,\ldots,j_n,k}$ for some countable index set I_J, and

$$\begin{aligned} E_J &= U_J \setminus \big(U_{j_1,\ldots,(j_n-1)} \cup \cdots \cup U_1\big) \\ &= \Big(\bigcup_{k\in I_J} U_{J,k}\Big) \setminus \big(U_{j_1,\ldots,(j_n-1)} \cup \cdots \cup U_1\big) \\ &= \Big[U_{J,1} \setminus \big(U_{j_1,\ldots,(j_n-1)} \cup \cdots \cup U_1\big)\Big] \cup \Big[U_{J,2} \setminus \big(U_{j_1,\ldots,(j_n-1)} \cup \cdots \cup U_1\big)\Big] \cup \cdots \\ &= U_{J,1} \setminus \big(U_{j_1,\ldots,(j_n-1)} \cup \cdots \cup U_1\big) \cup \end{aligned}$$

$$
\begin{aligned}
&\Big[\big(U_{J,2} \cap U_{J,1}^c \cap U_{j_1,\dots,(j_n-1)}^c \cap \cdots \cap U_1^c\big) \cup \\
&\qquad\qquad \big(U_{J,2} \cap U_{J,1} \cap U_{j_1,\dots,(j_n-1)}^c \cap \cdots \cap U_1^c\big)\Big] \cup \\
&\Big[\big(U_{J,3} \cap U_{J,2}^c \cap U_{J,1}^c \cap U_{j_1,\dots,(j_n-1)}^c \cap \cdots \cap U_1^c\big) \cup \\
&\qquad\qquad \big(U_{J,3} \cap U_{J,2} \cap U_{J,1} \cap U_{j_1,\dots,(j_n-1)}^c \cap \cdots \cap U_1^c\big) \cup \\
&\qquad\qquad \big(U_{J,3} \cap U_{J,2} \cap U_{J,1}^c \cap U_{j_1,\dots,(j_n-1)}^c \cap \cdots \cap U_1^c\big) \cup \cdots \\
&\qquad\qquad \big(U_{J,3} \cap U_{J,2}^c \cap U_{J,1} \cap U_{j_1,\dots,(j_n-1)}^c \cap \cdots \cap U_1^c\big)\Big] \cup \cdots \\
&= E_{J,1} \cup E_{J,2} \cup E_{J,3} \cup \cdots \\
&= \bigcup_{k \in I_J} E_{J,k}.
\end{aligned}
$$

That is, we split, for example, $U_{J,2} \setminus \big(U_{j_1,\dots,(j_n-1)} \cup \cdots \cup U_1\big)$ into two parts by intersecting with $X = U_{J,1}^c \cup U_{J,1}$. Then, the term that includes $U_{J,1}^c$ is exactly $E_{J,2}$, whereas the term that includes $U_{J,1}$ is a subset of $E_{J,1}$ and can therefore be ignored when taking the union. More generally, we split $U_{J,k} \setminus \big(U_{j_1,\dots,(j_n-1)} \cup \cdots \cup U_1\big)$ into 2^{k-1} sets by intersecting with

$$X = \bigcup \left\{ V_{J,k-1} \cap \cdots \cap V_{J,1} : V_{J,\ell} = U_{J,\ell}^c \text{ or } V_{J,\ell} = U_{J,\ell}, \text{ for } \ell = 1, \dots, k-1 \right\}.$$

Intersection with $U_{J,(k-1)}^c \cap \cdots \cap U_{J,1}^c$ gives us $E_{J,k}$. Of the remaining $2^{k-1} - 1$ terms, $2^{k-\ell}$ of them are subsets of $E_{J,(\ell-1)}$ for $\ell = 2, \dots, k$. As such, we can again exclude them from the overall union.

Since any set in $\mathcal{E}_n$ decomposes uniquely into a union of sets from $\mathcal{E}_{n+1}$, and since the sets in $\mathcal{E}_{n+1}$ are disjoint by construction, it follows from induction that $\{\mathcal{E}_n\}_{n=1}^\infty$ is a nested partition. Finally, by construction, for any $E_J \in \mathcal{E}_n$,

$$E_J \subseteq U_J \subseteq B(1/n, x_J)$$

for some $x_J \in D$. Thus, $\mathrm{Res}(\mathcal{E}_n) \leq 2/n$, so $\mathrm{Res}(\mathcal{E}_n)$ goes to zero.

□

Remark 3.8. Note that if X is compact, then each open cover $\mathcal{U}_n$ in the above construction can be replaced with a finite subcover, giving us a nested sequence $\{\mathcal{E}_n\}_{n=1}^\infty$ of finite partitions.

Theorem 3.9. *Let $F \in L_\mu^p(X, \mathcal{P}_p(Y))$. Then, there is a sequence $\{F_n\}_{n=1}^\infty$ of countably simple L^p Markov kernels that converges uniformly to F in the Wasserstein metric and that is weakly Cauchy in the L^p metric.*

Proof. Let $F \in L_\mu^p(X, \mathcal{P}_p(Y))$. Since $(\mathcal{P}_p(Y), W_{p,d_Y})$ is Polish, we can take a nested partition $\{\mathcal{E}_n\}_{n=1}^\infty$, as in Lemma (3.7). For each n, we construct a countably simple Markov kernel F_n as follows: Choose some enumeration $\{E_j^n\}_{j=1}^\infty$ for $\mathcal{E}_n$, and set $A_j^n \coloneqq F^{-1}\big(E_j^n\big)$. Note that $\{A_j^n\}_{j=1}^\infty$ is a pairwise disjoint collection of measurable

sets whose union equals X. For each j, pick some $x_j^n \in A_j^n$, and note that, by construction, there is $U_{j_1,\dots,j_n} \subseteq \mathcal{P}_p(Y)$ such that for all $m < n$,

$$F(A_j^n) \subseteq E_j^n \subseteq U_{j_1,\dots,j_n}^n \subseteq B\big(1/n, \nu_{j_1,\dots,j_m,\dots,j_n}\big) \subseteq B\big(1/m, \nu_{j_1,\dots,j_m}\big),$$

where the elements ν_J for multi-indices J are elements of some dense set $D \subseteq \mathcal{P}_p(Y)$, again as in Lemma (3.7). Then, define

$$F_n \coloneqq \sum_{j=1}^{\infty} F\big(x_j^n\big)\mathbf{1}_{A_j^n}.$$

By construction, $\{F_n\}_{n=1}^{\infty}$ converges uniformly in the Wasserstein metric to F. Indeed, given $\varepsilon > 0$, we can take $N > 2/\varepsilon$. Then, for any $x \in X$ and for any $n \geq N$, there is a unique j such that $x \in A_j^n$, and

$$\begin{aligned} W_{p,d_Y}\big(F_n(x), F(x)\big) &= W_{p,d_Y}\big(F(x_j^n), F(x)\big) \\ &\leq W_{p,d_Y}\big(F(x_j^n), \nu_{j_1,\dots,j_N}\big) + W_{p,d_Y}\big(\nu_{j_1,\dots,j_N}, F(x)\big) \\ &< 2/N < \varepsilon. \end{aligned}$$

Each F_n is also L^p integrable. To see this, fix some $\nu_0 \in \mathcal{P}_p(Y)$, and set $A \coloneqq F^{-1}\left(W(1,\nu_0)\right)$. Then,

$$\begin{aligned} &\int_X W_{p,d_Y}^p\big(\nu_0, F_n(x)\big)\mu(dx) \\ &\qquad \leq \int_X \left(W_{p,d_Y}\big(\nu_0, F(x)\big) + W_{p,d_Y}\big(F(x), F_n(x)\big)\right)^p \mu(dx) \\ &\qquad \leq \int_X \left(W_{p,d_Y}\big(\nu_0, F(x)\big) + \frac{1}{n}\right)^p \mu(dx) \\ &\qquad = \sum_{k=0}^{p} \binom{p}{k} \left(\frac{1}{n}\right)^{p-k} \left(\int_{X\setminus A} W_{p,d_Y}^k\big(\nu_0, F(x)\big)\mu(dx) \right. \\ &\qquad\qquad\qquad \left. + \int_A W_{p,d_Y}^k\big(\nu_0, F(x)\big)\mu(dx)\right) \\ &\qquad \leq \sum_{k=0}^{p} \binom{p}{k} \left(\frac{1}{n}\right)^{p-k} \left(\int_{X\setminus A} W_{p,d_Y}^p\big(\nu_0, F(x)\big)\mu(dx) + \mu(A)\right) \\ &\qquad < \infty, \end{aligned}$$

as needed.

Finally, it remains to show that $\{F_n\}_{n=1}^{\infty}$ is weakly Cauchy in the L^p metric. This follows immediately from uniform convergence. Let $0 < \varepsilon < 1$, and take N sufficiently large so that for all x and all $n \geq N$, $W_{p,d_Y}\big(F_n(x), F(x)\big) < \varepsilon/2$. Then, for all $n, m \geq N$,

$$\int_X W_{p,d_Y}^p\big(F_n(x), F_m(x)\big)\mu(dx)$$

$$\leq \int_X \left(W_{p,d_Y}\left(F_n(x), F(x)\right) + W_{p,d_Y}\left(F(x), F_m(x)\right)\right)^p \mu(dx)$$
$$\leq \int_X \varepsilon^p \mu(dx) < \varepsilon.$$

□

Proposition 3.10. *Let $\{F_n\}_{n=1}^\infty$ be a sequence of countably simple, weakly Cauchy L^p Markov kernels that converges uniformly in the Wasserstein metric to some $F \in L^p_\mu(X, \mathcal{P}_p(Y))$. Then, the limit*

$$\nu_F := \lim_{n\to\infty} \int_X F_n(x)\mu(dx)$$

exists in $\mathcal{P}_p(Y)$, and if $\{G_n\}_{n=1}^\infty$ is any other such sequence, then $\nu_F = \nu_G$.

Proof. By Proposition 3.5, or, more specifically, by inequality (3) from above, being weakly Cauchy in the L^p metric implies that the sequences $\int_X F_n(x)\mu(dx)$ and $\int_X G_n(x)\mu(dx)$ are both Cauchy in the Wasserstein metric. Since $\mathcal{P}_p(Y)$ is complete, these sequences converge to ν_F and ν_G, respectively, and, in fact, they are equal since

$$\begin{aligned} W^p_{p,d_Y}(\nu_F, \nu_G) &= W^p_{p,d_Y}\left(\lim_{n\to\infty}\int_X F_n(X)\mu(dx), \lim_{n\to\infty}\int_X G_n(x)\mu(dx)\right) \\ &= \lim_{n\to\infty} W^p_{p,d_Y}\left(\int_X F_n(x)\mu(dx), \int_X G_n(x)\mu(dx)\right) \\ &\leq \lim_{n\to\infty}\int_X \left(W_{p,d_Y}\left(F_n(x), F(x)\right) + W_{p,d_Y}\left(F(x), G_n(x)\right)\right)^p \mu(dx) \\ &= 0, \end{aligned}$$

where the last equality follows from uniform convergence. □

We can now define the Markov-Lebesgue integral for general L^p Markov kernels.

Definition 3.11. Let $F \in L^p_\mu(X, \mathcal{P}_p(Y))$. Then, the **Markov-Lebesgue integral** of F with respect to μ is defined as

$$\int_X F(x)\mu(dx) := \lim_{n\to\infty}\int_X F_n(x)\mu(dx),$$

where $\{F_n\}_{n=1}^\infty$ is any sequence of countably simple, weakly Cauchy L^p Markov kernels converging uniformly to F in the Wasserstein metric. By Proposition 3.10, the Markov-Lebesgue integral is well-defined.

In the remainder of this subsection, we verify that the measure defined by the Markov-Lebesgue integral is equivalent to the measure defined by the iterated integral. Additionally, we prove a Fubini-type result.

Theorem 3.12. *Let $F \in L^p_\mu(X, \mathcal{P}_p(Y))$, and let $\phi \in C_b(Y)$. Then,*

$$\int_Y \phi(y)\left(\int_X F(x)\mu(dx)\right)(dy) = \int_X \left(\int_Y \phi(y)F_x(dy)\right)\mu(dx).$$

That is, the probability measure defined by the Markov-Lebesgue integral is equal to the measure defined by the iterated integral.

Proof. Let $F \in L^p_\mu(X, \mathcal{P}_p(Y))$, and let $\{F_n\}_{n=1}^\infty$ be a sequence of countably simply, weakly Cauchy L^p Markov kernels converging uniformly to F. Proposition 3.3 shows that the result holds for countably simple L^p Markov kernels. Thus,

$$\begin{aligned}\int_Y \phi(y)\left(\int_X F(x)\mu(dx)\right)(dy) &= \int_Y \phi(y)\left(\int_X \lim_{n\to\infty} F_n(x)\mu(dx)\right)(dy)\\ &= \lim_{n\to\infty}\int_Y \phi(y)\left(\int_X F_n(x)\mu(dx)\right)(dy)\\ &= \lim_{n\to\infty}\int_X \left(\int_Y \phi(y)F_n(dy|x)\right)\mu(dx)\\ &= \int_X \left(\int_Y \phi(y)F_x(dy)\right)\mu(dx).\end{aligned}$$

□

Theorem 3.13 (**Fubini**). *Let $F \in L^p_{\mu_1\otimes\mu_2}(X_1\times X_2, \mathcal{P}_p(Y_1\times Y_2))$. Then, $F(x_1,\cdot) \in L^p_{\mu_2}(X_2, \mathcal{P}_p(Y_1\times Y_2))$ and $F(\cdot, x_2) \in L^p_{\mu_1}(X_1, \mathcal{P}_p(Y_1\times Y_2))$. Further, the maps*

$$x_1 \mapsto \int_{X_2} F(x_1,x_2)\mu_2(dx_2) \ \text{ and } \ x_2 \mapsto \int_{X_1} F(x_1,x_2)\mu_1(dx_1)$$

are in $L^p_{\mu_1}(X_1, \mathcal{P}_p(Y_1\times Y_1))$ and $L^p_{\mu_2}(X_2, \mathcal{P}_p(Y_1\times Y_2))$, respectively, and

$$\begin{aligned}&\int_{X_1\times X_2} F(x_1,x_2)(\mu_1\otimes\mu_2)(dx_1,dx_2)\\ &\qquad= \int_{X_1}\left(\int_{X_2} F(x_1,x_2)\mu_2(dx_2)\right)\mu_1(dx_1)\\ &\qquad= \int_{X_2}\left(\int_{X_1} F(x_1,x_2)\mu_1(dx_1)\right)\mu_2(dx_2).\end{aligned}$$

Proof. First, fix $x_0 \in X_1$ and $z_0 = (y_{01}, y_{02}) \in Y_1\times Y_2$, and let $z = (y_1, y_2)$. Recall that, by assumption, we equip $Y_1\times Y_2$ with the metric $d^p_{Y_1Y_2} = d^p_{Y_1} + d^p_{Y_2}$. Then,

$$\begin{aligned}&\int_{X_2}\left(\int_{Y_1\times Y_2} d^p_{Y_1Y_2}(z_0,z)F(dz|x_0,x_2)\right)\mu_2(dx_2)\\ &\quad\le \int_{X_1}\left(\int_{X_2}\left(\int_{Y_1\times Y_2} d^p_{Y_1Y_2}(z_0,z)F(dz|x_1,x_2)\right)\mu_2(dx_2)\right)\mu_1(dx_1)\\ &\quad= \int_{X_1\times X_2}\left(\int_{Y_1\times Y_2} d^p_{Y_1Y_2}(z_0,z)F(dz|x_1,x_2)\right)(\mu_1\otimes\mu_2)(dx_1,dx_2)\\ &\quad< \infty.\end{aligned}$$

Thus, $F(x_0,\cdot)$ is in $L^p_{\mu_2}(X_2, \mathcal{P}_p(Y_1\times Y_2))$, and similarly, $F(\cdot,x_2) \in L^p_{\mu_1}(X_1, \mathcal{P}_p(Y_1\times Y_2))$ for any $x_2 \in X_2$. Further, since the Markov-Lebesgue integral is equivalent to

the iterated integral, we have

$$\begin{aligned}
&\int_{X_1}\left(\int_{Y_1\times Y_2} d^p_{Y_1Y_2}(z_0,z)\left(\int_{X_2} F(x_1,x_2)\mu_2(dx_2)\right)(dz)\right)\mu_1(dx_1)\\
&\quad=\int_{X_1}\left(\int_{X_2}\left(\int_{Y_1\times Y_2} d^p_{Y_1Y_2}(z_0,z)F(dz|x_1,x_2)\right)\mu_2(dx_2)\right)\mu_1(dx_1)\\
&\quad<\infty.
\end{aligned}$$

Thus, the map $x_1 \mapsto \int_{X_1} F(x_1,x_2)\mu_(dx_2)$ is in $L^p_{\mu_1}(X_1,\mathcal{P}_p(Y_1\times Y_2))$. Likewise, the map $x_2 \mapsto \int_{X_2} F(x_1,x_2)\mu_1(dx_1)$ is in $L^p_{\mu_2}(X_2,\mathcal{P}_p(Y_1\times Y_2))$.

Finally, let $\phi \in C_b(Y_1\times Y_2)$. Again by the equivalence between the Markov-Lebesgue integral and the iterated integral, we have

$$\begin{aligned}
&\int_{Y_1\times Y_2}\phi(y_1,y_2)I_{\mu_1\otimes_\nabla\mu_2}(F)(dy_1,dy_2)\\
&\quad=\int_{X_1\times X_2}\left(\int_{Y_1\times Y_2}\phi(y_1,y_2)F_{x_1,x_2}(dy_1,dy_2)\right)(\mu_1\otimes\mu_2)(dx_1,dx_2)\\
&\quad=\int_{X_1}\left(\int_{X_2}\left(\int_{Y_1\times Y_2}\phi(y_1,y_2)F_{x_1,x_2}(dy_1,dy_2)\right)\mu_2(dx_2)\right)\mu_1(dx_1)\\
&\quad=\int_{X_1}\left(\int_{Y_1\times Y_2}\phi(y_1,y_2)I_{\mu_2}(F_{x_1})(dy_1,dy_2)\right)\mu(dx_1)\\
&\quad=\int_{Y_1\times Y_2}\phi(y_1,y_2)\left(\int_{X_1} I_{\mu_2}(F_{x_1})\mu(dx_1)\right)(dy_1,dy_2).
\end{aligned}$$

Thus,

$$\int_{X_1\times X_2} F(x_1,x_2)(\mu_1\otimes\mu_2)(dx_1,dx_2)=\int_{X_1}\left(\int_{X_2} F(x_1,x_2)\mu_2(dx_2)\right)\mu_1(dx_1).$$

A similar computation shows that we can integrate in the other order. That is, since $\mu_1\otimes\mu_2$ is independent, we can flip the order of integration on line three of the above computation after writing the Markov-Lebesgue integral as the iterated integral. □

Remark 3.14. If $F \in L^p_{\mu_1}(X_1,\mathcal{P}_p(Y_1))$ and $G \in L^p_{\mu_2}(X_2,\mathcal{P}_p(Y_2))$, then the Markov kernel

$$F\otimes G : X_1\times X_2 \to \mathcal{P}_p(Y_1\times Y_2)$$

defined by

$$(F\otimes G)(x_1,x_2)=F(x_1)\otimes G(x_2)$$

is in $L^p_{\mu_1\otimes\mu_2}(X_1\times X_2,\mathcal{P}_p(Y_1\times Y_2))$. Indeed, fix $z_0=(y_{01},y_{02})\in Y_1\times Y_2$, and set $z=(y_1,y_2)$. Then, the integral splits into two parts:

$$\begin{aligned}
&\int_{X_1\times X_2}\left(\int_{Y_1\times Y_2} d^p_{Y_1Y_2}(z_0,z)(F\otimes G)(dz|x_1,x_2)\right)(\mu_1\otimes\mu_2)(dx_1,dx_2)\\
&\quad=\int_{X_1}\left(\int_{Y_1} d^p_{Y_1}(y_{01},y_1)F_{x_1}(dy_1)\right)\mu_1(dx_1)
\end{aligned}$$

$$+ \int_{X_2} \left(\int_{Y_1} d^p_{Y_2}(y_{02}, y_2) F_{x_2}(dy_2) \right) \mu_2(dx_2)$$
$$< \infty.$$

Thus, the Fubini theorem for Markov kernels applies to $F \otimes G$. Similarly, if $F \in L^p_\mu(X, \mathcal{P}_p(Y))$, then $F^{\otimes n} : X^n \to \mathcal{P}_p(Y^n)$ is in $L^p_{\mu^{\otimes n}}(X^n, \mathcal{P}_p(Y^n))$, and the map $F^{\otimes n} \circ \mathrm{copy}^n_X : X \to \mathcal{P}_p(Y^n)$ is in $L^p_\mu(X, \mathcal{P}_p(Y^n))$.

3.2. **Disintegration.** We now present an altered version of the Disintegration Theorem, which we call the L^p Disintegration Theorem. Using the theory we've developed for the Markov-Lebesgue integral, we show that any joint distribution can be recovered as the Markov-Lebesgue integral of a Markov kernel, where, of course, we view Markov kernels as integrable maps taking values in a Wasserstein space.

Definition 3.15. Let $\gamma \in \mathcal{P}_p(X \times Y)$ with marginals $\gamma^1 = \mu$ and $\gamma^2 = \nu$. Then, a Markov kernel $F \in L^p_\mu(X, \mathcal{P}_p(Y))$ is a called a **disintegration** of γ if for all $\phi \in C_b(X \times Y)$,

$$\int_{X \times Y} \phi(x, y) \gamma(dx, dy) = \int_X \left(\int_Y \phi(x, y) F_x(dy) \right) \mu(dx).$$

In [59], Villani writes $\int_X \big(\delta_x \otimes F_x\big)\mu(dx)$ for the measure defined by the iterated integral above. In Theorem 3.16 below, we show that this notation can be taken literally as the Markov-Lebesgue integral.

Theorem 3.16 (**L$^{\mathbf{p}}$ Disintegration Theorem**). *Let $\gamma \in (\mathcal{P}_p(X \times Y), W_{p,d_{XY}})$ with $\gamma^1 = \mu$ and $\gamma^2 = \nu$. Then, there exist almost surely unique Markov kernels $F \in L^p_\mu(X, \mathcal{P}_p(Y))$ and $G \in L^p_\nu(Y, \mathcal{P}_p(X))$ with*

$$\int_X \big(\delta_x \otimes F_x\big)\mu(dx) = \gamma = \int_Y \big(G_y \otimes \delta_y\big)\nu(dy), \tag{4}$$

where the above is now an equality of Markov-Lebesgue integrals.

Proof. We follow the proof of the ordinary Disintegration Theorem for the bulk of the argument, as presented in, for example, Theorem 10.4.5 of Bogachev [9]. Let $\gamma \in \mathcal{P}_p(X \times Y)$ with $\gamma^1 = \mu$ and $\gamma^2 = \nu$. Since the spaces X, Y, and $X \times Y$ are all standard Borel, there exist countable collections $\mathcal{A} = \{A_j\}_{j=1}^\infty$ and $\mathcal{B} = \{B_j\}_{j=1}^\infty$ generating Σ_X and Σ_Y, respectively, and, further, $\mathcal{E} = \{A_j \times B_k\}_{j,k=1}^\infty$ generates $\Sigma_{X \times Y} = \sigma(\Sigma_X \times \Sigma_Y)$. By regularity and tightness of measures on standard Borel spaces, we have that for any $B_j \in \mathcal{B}$, there exist sets B_{jk} and K_{jk} in Σ_X with K_{jk} compact such that

$$B_{jk} \subseteq K_{jk} \subseteq B_j \quad \text{and} \quad \nu(B_j \setminus B_{jk}) < 1/k.$$

The collection $\mathcal{K}$ of all sets K_{jk} forms what is called a compact class, and we say that $\mathcal{K}$ approximates μ on $\mathcal{B}$ with respect to Σ_Y.

Now, let $\mathcal{C} \subseteq \Sigma_Y$ denote the countable subalgebra generated by all sets of the form B_j and B_{jk} by taking complements as well as pairwise unions and intersections. Given $C \in \mathcal{C}$, the measure $\gamma(- \times C) : \Sigma_X \to [0, 1]$ is absolutely continuous with

respect to μ, since if $\mu(A) = 0$, then $\gamma(A \times C) \leq \gamma(A \times Y) = \mu(A) = 0$. As such, by the Radon-Nikodym Theorem, there exists a function $F(C|-) : X \to \mathbb{R}$ in $L^1(\mu)$ such that

$$\gamma(A \times C) = \int_A F(C|x)\mu(dx) \tag{5}$$

for any $A \in \Sigma_X$. Further, given disjoint measurable sets C_1 and C_2 in $\mathcal{C}$, we have

$$\begin{aligned}\int_A F(C_1 \cup C_2|x)\mu(dx) &= \gamma(A \times (C_1 \cup C_2)) \\ &= \gamma(A \times C_1 \cup A \times C_2) \\ &= \gamma(A \times C_1) + \gamma(A \times C_2) \\ &= \int_A F(C_1|x) + F(C_2|x)\mu(dx).\end{aligned}$$

Hence, $F(C_1 \cup C_2|x) = F(C_1|x) + F(C_2|x)$ for μ-a.e. $x \in X$. Let $N_{jk} \subseteq X$ denote the set of all x where this equality fails. Then, the union N_0 of all N_{jk} for $1 \leq j, k < \infty$ has measure zero, since it is a countable collection of sets of measure zero. Thus, for μ-a.e. $x \in X$, $F(-|x) : \mathcal{C} \to [0, 1]$ is finitely additive.

Now, define

$$f_j(x) \coloneqq \sup_k F(B_{jk}|x).$$

We'll show that for μ-a.e. $x \in X$, $F(B_j|x) = f_j(x)$, which will show that $\mathcal{K}$ is a compact class approximating F on $\mathcal{C}$ with respect to Σ_Y. First, note that f_j is measurable with respect to Σ_X, since each $F(B_{jk}|-)$ is in $L^1(\mu)$. By definition, $F(B_{jk}|x) \leq f_j(x)$, which gives us

$$\nu(B_{jk}) = \gamma(X \times B_{jk}) = \int_X F(B_{jk}|x)\mu(dx) \leq \int_X f_j(x)\mu(dx).$$

Also, since $B_{jk} \subseteq B_j$, there is a set $M_{jk} \subseteq X$ of μ-measure zero such that $F(B_{jk}|x) \leq F(B_j|x)$ for $x \notin M_{jk}$. Thus, for all $x \notin M_0 \coloneqq \bigcup_{j,k} M_{jk}$, $f_j(x) \leq F(B_j|x)$, and

$$\nu(B_j) = \sup_k \nu(B_{jk}) \leq \int_X f_j(x)\mu(dx) \leq \int_X F(B_j|x)\mu(dx) = \nu(B_j).$$

Thus, $f_j(x) = F(B_j|x)$ for all x outside of some set M_1 of μ-measure zero. It follows that for all $x \notin M_0 \cup M_1$, $\mathcal{K}$ is a compact class approximating $F(-|x)$. By Theorem 1.4.3 and Remark 1.4.7 in [9], F is therefore countably additive on $\mathcal{C}$. By Carathéodory's Extension Theorem, $F(-|x)$ extends uniquely to a measure on Σ_Y, and from (5), $F(-|x)$ is a probability measure.

This construction gives us an almost surely unique Markov kernel $F : \Sigma_Y \times X \to [0, 1]$ with the desired properties. Indeed, for any $C \in \mathcal{C}$, $x \mapsto F(C|x)$ is measurable with respect to Σ_X. Let $\mathcal{C}_0$ denote the collection of sets for which this is true. Then, for an increasing sequence of sets $\{C_n\}_{n=1}^\infty$, $F(\cup_{n=1}^\infty C_n)$ is the pointwise limit of Borel functions and is therefore measurable. Hence, $\mathcal{C}_0$ is closed under increasing unions, and similarly, it is closed under decreasing intersections. As such, $\mathcal{C}_0$ is a monotone class, and by the Monotone Class Theorem (e.g., Theorem 1.9.3 in [9]),

we have $\mathcal{C}_0 = m(\mathcal{C}_0) = \sigma(\mathcal{C}_0) = \sigma(\mathcal{C}) = \Sigma_Y$, where $m(\mathcal{C}_0)$ is the monotone class generated by $\mathcal{C}_0$. Thus, $x \mapsto F(B|x)$ is measurable for any $B \in \Sigma_Y$.

Additionally, for any $A \in \Sigma_X$ and $C \in \mathcal{C}$, (5) is true by construction, and for fixed $A \in \Sigma_X$, the class of sets $\mathcal{C}_1 \subseteq \sigma(\mathcal{C})$ for which (5) holds is monotone. Indeed, if a sequence $\{C_n\}_{n=1}^{\infty}$ increases to C, then $\lim_{n\to\infty} F(C_n|x) = F(C|x)$, and by continuity from below, together with the Dominated Convergence Theorem, we have

$$\gamma\left(A \times \cup_{n=1}^{\infty} C_n\right) = \int_X F\left(\cup_{n=1}^{\infty} C_n|x\right)\mu(dx).$$

Similarly, we have closure under intersections. $\mathcal{C}_1$ is therefore a monotone class, and $\mathcal{C}_1 = m(\mathcal{C}_1) = \sigma(\mathcal{C}) = \Sigma_Y$. Thus, (5) holds for all $A \times B \in \Sigma_X \times \Sigma_Y$. In particular, note that for all $B \in \Sigma_Y$,

$$\nu(B) = \gamma(X \times B) = \int_X F(B|x)\mu(dx). \tag{6}$$

Equalities (5) and (6) then imply that for all $\phi \in C_b(X \times Y)$ and all $\psi \in C_b(Y)$,

$$\int_{X\times Y} \phi(x,y)\gamma(dx,dy) = \int_X \left(\int_Y \phi(y)F(dy|x)\right)\mu(dx)$$

and

$$\int_Y \psi(y)\nu(dy) = \int_X \left(\int_Y \psi(y)F_x(dy)\right)\mu(dx).$$

This follows since it is enough to check on indicator functions, which is immediate.

The above therefore establishes the ordinary Disintegration Theorem. To re-interpret the result using the Markov-Lebesgue integral, note first that

$$\int_X \left(\int_Y d_Y^p(y_0,y)F(dy|x)\right)\mu(dx) = \int_Y d_Y^p(y_0,y)\nu(dy) < \infty.$$

That is, F has finite pth moment, and $F_x \in \mathcal{P}_p(Y)$ for μ-a.e. $x \in X$. By Theorem 2.25, we may equivalently take F to be an L^p integrable Borel map $F : X \to \mathcal{P}_p(Y)$. Additionally, note that the map $x \mapsto \delta_x$ is in $L^p_\mu(X, \mathcal{P}_p(X))$. By Remark 3.14, $\delta \otimes F$ is in $L^p_{\mu\otimes\mu}(X \times X, \mathcal{P}_p(X \times Y))$. Let $\mathrm{copy}_X : X \to X \times X$ denote the map given by $\mathrm{copy}_X(x) = (x,x)$. Then, the map $x \mapsto \delta_x \otimes F_x$ is equal to $(\delta \otimes F) \circ \mathrm{copy}_X$. Fix $z_0 = (x_0, y_0)$, and let $D \subseteq X \times X$ be the diagonal. Then,

$$\begin{aligned}
&\int_X \left(\int_{X\times Y} d_{XY}^p(z_0,z)\big(\delta_x \otimes F_x\big)(dz)\right)\mu(dx) \\
&\quad = \int_D \left(\int_{X\times Y} d_{XY}^p(z_0,z)(\delta_{x_1} \otimes F_{x_2})(dz)\right)(\mu\otimes\mu)(dx_1,dx_2) \\
&\quad \le \int_{X\times X} \left(\int_{X\times Y} d_{XY}^p(z_0,z)(\delta_{x_1} \otimes F_{x_2})(dz)\right)(\mu\otimes\mu)(dx_1,dx_2) \\
&\quad < \infty.
\end{aligned}$$

Thus, the map $(\delta \otimes F) \circ \mathrm{copy}_X$ is in $L^p_\mu(X, \mathcal{P}_p(X \times Y))$. As such, it is integrable, and by the equivalence of the Markov-Lebesgue integral and the iterated integral,

as stated in Theorem 3.12, we have that for any rectangle $A \times B \in \Sigma_X \times \Sigma_Y$,

$$\begin{aligned}
&\left(\int_X \big(\delta_x \otimes F_x\big)\mu(dx)\right)(A \times B) \\
&\qquad = \int_X \left(\int_{X\times Y} \mathbf{1}_{A\times B}(x_1, y_1)\big(\delta_x \otimes F_x\big)(dx_1, dy_1)\right)\mu(dx) \\
&\qquad = \int_X \left(\int_X \mathbf{1}_A(x_1)\delta_x(dx_1)\int_Y \mathbf{1}_B(y_1)F_x(dy_1)\right)\mu(dx) \\
&\qquad = \int_X \delta_x(A)F_x(B)\mu(dx) \\
&\qquad = \int_A F_x(B)\mu(dx) \\
&\qquad = \gamma(A \times B).
\end{aligned}$$

Since $X \times Y$ is standard Borel, this suffices to show equality on all of $\Sigma_{X\times Y}$.

Finally, an exactly symmetrical argument to everything presented above shows that there is also $G \in L^p_\nu(Y, \mathcal{P}_p(X))$ with

$$\gamma = \int_Y \big(G_y \otimes \delta_y\big)\nu(dy).$$

Additionally, similar reasoning shows that the equalities

$$\nu = \int_X F(x)\mu(dx) \quad \text{and} \quad \mu = \int_Y G(y)\nu(dy)$$

also hold when interpreted as Markov-Lebesgue integrals. □

To conclude this subsection, we recall one last definition, namely, a variant of the Wasserstein metric commonly called the adapted, or nested, Wasserstein metric, which we will use in later sections.

Definition 3.17. Let (X, d_X) and (Y, d_Y) be complete separable metric spaces. The **adapted Wasserstein metric of order p**

$$AW_p : \mathcal{P}_p(X \times Y) \times \mathcal{P}_p(X \times Y) \to [0, \infty)$$

is defined by

$$AW_p(\gamma_1, \gamma_2) = \left(\inf_{\pi:\Pi(\mu_1,\mu_2)} \int_{X^2} \Big(d^p_X(x_1, x_2) + W^p_{p,d_Y}\big(F_{x_1}, G_{x_2}\big)\Big)\,\pi(dx_1, dx_2)\right)^{1/p}$$

where $F : X \to \mathcal{P}_p(Y)$ and $G : X \to \mathcal{P}_p(Y)$ are almost surely unique disintegrations of γ_1 and γ_2, respectively.

Lemma 3.18. *Let $\gamma_1, \gamma_2 \in \mathcal{P}_p(X \times Y)$ be joint distributions. Then,*

$$W_{p,d_{XY}}(\gamma_1, \gamma_2) \le AW_p(\gamma_1, \gamma_2).$$

Proof. Let $\gamma_1, \gamma_2 \in \mathcal{P}_p(X \times Y)$. By the Disintegration Theorem, there exist μ_j-almost surely unique Markov kernels F and G such that $\gamma_1 = I_{\mu_1}(\delta_x \otimes_\nabla F_x)$ and $\gamma_2 = I_{\mu_2}(\delta_x \otimes_\nabla G_x)$, where $\mu_j = \gamma^1_j$ is the first marginal of γ_j for $j = 1, 2$. Let

$\sigma : X \times X \to \mathcal{P}_p(Y \times Y)$ be the optimal pairwise coupling between fibers F_{x_1} and G_{x_2} given by Theorem 2.23. Note that for any $\pi \in \Pi(\mu_1, \mu_2)$, the integral $I_\pi(\sigma) \in \Pi(\gamma_1, \gamma_2)$. Thus, letting $z_j = (x_j, y_j)$ for $j = 1, 2$, we have

$$\begin{aligned}
AW_p^p(\gamma_1, \gamma_2) &= \inf_{\pi:\Pi(\mu_1,\mu_2)} \int_{X^2} \left(d_X^p(x_1, x_2) + W_{p,d_Y}^p\left(F_{x_1}, G_{x_2}\right)\right) \pi(dx_1, dx_2) \\
&= \inf_{\pi:\Pi(\mu_1,\mu_2)} \int_{X^2} \Bigg(d_X^p(x_0, x_1) + \\
&\qquad\qquad \int_{Y^2} d_Y^p(y_0, y_1)\sigma_{x_1,x_2}(dy_1, dy_2)\Bigg) \pi(dx_1, dx_2) \\
&= \inf_{\pi:\Pi(\mu_1,\mu_2)} \int_{(X\times Y)^2} d_{XY}^p(z_1, z_2) I_\pi(\sigma)(dz_1, dz_2) \\
&\geq \inf_{\omega:\Pi(\gamma_1,\gamma_2)} \int_{(X\times Y)^2} d_{XY}^p(z_1, z_2)\omega(dz_1, dz_2) \\
&= W_{p,d_{XY}}^p(\gamma_1, \gamma_2).
\end{aligned}$$

□

4. p-Kantorovich Monads

In this section, we prove that the general Wasserstein space construction defines a monad on a certain category of metric spaces. Categorical language is not strictly necessary for our results in Section 5. However, by showing that our main objects of study have a nice categorical structure, we establish a basis from which to develop further connections between the structural questions of categorical probability and probabilistic programming semantics and the quantitative questions of statistical inference. More prosaically, to make our general theory work, we need to show various preliminary results, which, taken together, essentially amount to a verification of the existence of a monadic structure for Wasserstein spaces. The case of $p = 1$ is well-known, and the construction is called the Kantorovich monad; see, for example, [27] or [57]. Thus, our contribution here is simply to show that, in fact, we can take $1 \leq p < \infty$. Naturally, we call the resulting construction the Kantorovich monad of order p, or the p-Kantorovich monad.

Any reader uninterested in the categorical framework can primarily pay attention to Lemmas 4.3 and 4.6, which are stated without categorical language. The reader will also need to acquire some of the notation defined throughout this section, specifically in definitions 4.5 and 4.8.

Definition 4.1. Let $\mathcal{C}$ be a category. A **monad** on $\mathcal{C}$ is an endofunctor $T : \mathcal{C} \to \mathcal{C}$ together with natural transformations $\eta : \mathrm{id}_{\mathcal{C}} \to T$ and $m : T^2 \Rightarrow T$, called the unit

and multiplication maps, respectively, such that the following diagrams commute:

$$\begin{tikzcd} T \arrow[r, Rightarrow, "\eta T"] \arrow[dr, Rightarrow, "\mathrm{id}_T"'] & T^2 \arrow[d, Rightarrow, "m"] & T \arrow[l, Rightarrow, "T\eta"'] \arrow[dl, Rightarrow, "\mathrm{id}_T"] \\ & T & \end{tikzcd} \qquad \begin{tikzcd} T^3 \arrow[r, Rightarrow, "mT"] \arrow[d, Rightarrow, "Tm"'] & T^2 \arrow[d, Rightarrow, "m"] \\ T^2 \arrow[r, Rightarrow, "m"'] & T \end{tikzcd}$$

(We define the natural transformations in the above diagrams in Proposition 4.9.)

In a phrase, we can say that a monad is essentially a monoid object in a category of endofunctors. That is, the multiplication map $m : T^2 \Rightarrow T$ is analogous to the binary operation in a monoid, and the unit map is analogous to the identity element.

Definition 4.2. Let **Met** denote the category of complete and separable metric spaces (X, d_X) where morphisms are given by Lipschitz functions $f : X \to Y$. That is, $f \in \mathrm{Hom}(X, Y)$ if there exists a constant $K > 0$ such that

$$d_Y\big(f(x_1), f(x_2)\big) \leq K d_X(x_1, x_2)$$

for all $x_1, x_2 \in X$. For $1 \leq p < \infty$, we define $(\mathbf{Met}_p, \otimes, I)$ to be the category **Met** equipped with the following monoidal structure: The tensor product $X \otimes Y$ is the set $X \times Y$ equipped with the product metric of order p:

$$(d_X \otimes d_Y)\big((x_1, y_1), (x_2, y_2)\big) = \big(d_X^p(x_1, x_2) + d_Y^p(y_1, y_2)\big)^{1/p}.$$

As before, we denote this metric by d_{XY}. Note that the unit object I, given by the one point metric space $\{\mathrm{pt}\}$, is terminal. Thus, $(\mathbf{Met}_p, \otimes, I)$ is semicartesian, or equivalently, a monoidal category with projections.

Lemma 4.3. *Let (X, d_X) and (Y, d_Y) be complete separable metric spaces, and let $f : X \to Y$ be Lipschitz. Then, the pushforward operation given by*

$$(f_\sharp \mu)(E) = \mu(f^{-1}(E))$$

defines a Lipschitz map

$$f_\sharp : \mathcal{P}_p(X) \to \mathcal{P}_p(Y),$$

where $\mathcal{P}_p(X)$ and $\mathcal{P}_p(Y)$ are equipped with their respective Wasserstein metrics.

Proof. First, we'll show that pushforwards by Lipschitz functions preserve finite moments. Let $K > 0$ be a Lipschitz constant for f, and let $\mu \in \mathcal{P}_p(X)$. Let $y_0 = f(x_0)$ for some $x_0 \in X$. Then,

$$\begin{aligned} \int_Y d_Y^p(y_0, y)(f_\sharp \mu)(dy) &= \int_X d_Y^p\big(f(x_0), f(x)\big)\mu(dx) \\ &\leq K^p \int_X d_X^p(x_0, x)\mu(dx) \\ &< \infty. \end{aligned}$$

Thus, $\mathcal{P}_p(f)(\mu) = f_\sharp\mu \in \mathcal{P}_p(Y)$. To see that $\mathcal{P}_p(f)$ is itself Lipschitz, we use Kantorovich duality. Set $c_1 \coloneqq d_Y^p$ and $c_2 \coloneqq K^p d_X^p$. It is straightforward to check that if $(\phi, \psi) \in \Phi_{c_1}(f_\sharp\mu, f_\sharp\nu)$, then $(\phi \circ f, \psi \circ f) \in \Phi_{c_2}(\mu, \nu)$. As such,

$$\begin{aligned} W^p_{p,d_Y}\big(f_\sharp\mu, f_\sharp\nu\big) &= \sup_{(\phi,\psi):\Phi_{c_1}} \int_Y \phi(y)(f_\sharp\mu)(dy) + \int_Y \psi(y)(f_\sharp\nu)(dy) \\ &= \sup_{(\phi,\psi):\Phi_{c_1}} \int_X (\phi \circ f)(x)\mu(dx) + \int_X (\psi \circ f)(x)\nu(dx) \\ &\leq \sup_{(\xi,\zeta):\Phi_{c_2}} \int_X \xi(x)\mu(dx) + \int_X \zeta(x)\nu(dx) \\ &= K^p W^p_{p,d_X}(\mu, \nu). \end{aligned}$$

□

Proposition 4.4. *The following assignments define an endofunctor*

$$\mathcal{P}_p : \mathbf{Met} \to \mathbf{Met}$$

on the category **Met***: On objects, $\mathcal{P}_p$ maps (X, d_X) to $(\mathcal{P}_p(X), W_{p,d_X})$. Further, given a Lipschitz function $f : X \to Y$,*

$$\mathcal{P}_p(f) : \mathcal{P}_p(X) \to \mathcal{P}_p(Y)$$

is defined as the pushforward: $\mathcal{P}_p(f)(\mu) = f_\sharp\mu$.

Proof. By Theorem 2.10, the Wasserstein space $(\mathcal{P}_p, W_{p,d_X})$ is complete and separable whenever (X, d_X) is. Further, Lemma 4.3 shows that $\mathcal{P}_p(f)$ is Lipschitz whenever f is. Hence, all that is left to show is that $\mathcal{P}_p(\mathrm{id}_X) = \mathrm{id}_{\mathcal{P}_p(X)}$ and that $\mathcal{P}_p(g \circ f) = \mathcal{P}_p(g) \circ \mathcal{P}_p(f)$. The first identity is immediate; the second is straightforward:

$$\begin{aligned} \mathcal{P}_p(g \circ f)(\mu)(E) &= (g \circ f)_\sharp(\mu)(E) \\ &= \mu\big((g \circ f)^{-1}(E)\big) \\ &= \mu\big((f^{-1} \circ g^{-1})(E)\big) \\ &= (f_\sharp\mu)(g^{-1}(E)) \\ &= g_\sharp(f_\sharp\mu)(E) \\ &= (\mathcal{P}_p(g) \circ \mathcal{P}_p(f))(E). \end{aligned}$$

□

Definition 4.5. Let (X, d_X) be Polish. Set $\mathcal{P}_p^2 \coloneqq \mathcal{P}_p \circ \mathcal{P}_p$, and let $P \in \mathcal{P}_p^2(X)$. The **expectation, or multiplication, map** $E_X : \mathcal{P}_p^2(X) \to \mathcal{P}_p(X)$ is defined on test functions $\phi \in C_b(X)$ by

$$\int_X \phi(x) E_X(P)(dx) = \int_{\mathcal{P}_p(X)} \left(\int_X \phi(x)\mu(dx) \right) P(d\mu).$$

In particular, this gives

$$E_X(P)(A) = \int_{\mathcal{P}_p(X)} \mathrm{ev}_A(\mu) P(d\mu)$$

for any $A \in \Sigma_X$. The reason for calling E_X the expectation map is that we can informally think of it as being of the form

$$E_X(P) = \int_{\mathcal{P}_p(X)} \mu P(d\mu).$$

As noted previously, we can likely extend the Markov-Lebesgue integral to a more general theory of integration (against probability measures) for functions taking values in convex Polish metric spaces. If so, the equation above could then be taken literally.

Lemma 4.6. *Let (X, d_X) be Polish. Equip $\mathcal{P}_p(X)$ with the usual Wasserstein metric, and equip $\mathcal{P}_p^2(X)$ with the iterated Wasserstein metric $\mathcal{W}_p := W_{p,W_{p,d_X}}$. Then, the multiplication map $E_X : \mathcal{P}_p^2(X) \to \mathcal{P}_p(X)$ is Lipschitz with Lipschitz constant equal to one.*

Proof. First, we verify that $E_X(P)$ is indeed in $\mathcal{P}_p(X)$ whenever $P \in \mathcal{P}_p^2(X)$. Recall that the product measure $\delta_{x_0} \otimes_\nabla \mu$ is the unique optimal coupling between a Dirac mass $\delta_{x_0} \in \mathcal{P}_p(X)$ and any $\mu \in \mathcal{P}_p(X)$. Then, since P has finite pth moment, we have

$$\begin{aligned}
&\int_X d_X^p(x_0, x) E_X(P)(dx) \\
&\quad = \int_{\mathcal{P}_p(X)} \left(\int_X d_X^p(x_0, x) \mu(dx) \right) P(d\mu) \\
&\quad = \int_{\mathcal{P}_p(X)} \left(\int_X \left(\int_X d_X^p(x_1, x_2) \delta_{x_0}(dx) \right) \mu(dx_2) \right) P(d\mu) \\
&\quad = \int_{\mathcal{P}_p(X)} \left(\int_{X \times X} d_X^p(x_1, x_2) (\delta_{x_0} \otimes \mu)(dx_1, dx_2) \right) P(d\mu) \\
&\quad = \int_{\mathcal{P}_p(X)} W_{p,d_X}^p(\delta_{x_0}, \mu) P(d\mu) \\
&\quad < \infty,
\end{aligned}$$

as needed.

Next, we'll show that E_X is Lipschitz. Equip $\mathcal{P}_p(X) \times \mathcal{P}_p(X)$ with the product metric $\left(W_{p,d_X} \otimes W_{p,d_X}\right)^p = W_{p,d_X}^p + W_{p,d_X}^p$, and equip $\mathcal{P}_p(X \times X)$ with the metric $W_{p,d_{XX}}$. Let $\mathrm{Cl}(\mathcal{P}_p(X \times X))$ denote the closed subsets of $\mathcal{P}_p(X \times X)$. Define $\Pi_o : \mathcal{P}_p(X) \times \mathcal{P}_p(X) \to \mathrm{Cl}(\mathcal{P}_p(X \times X))$ to be the mapping $(\mu, \nu) \mapsto \Pi_o(\mu, \nu)$. By the Measurable Selection Theorem (Theorem 2.15), there exists a Borel map $F : \mathcal{P}_p(X) \times \mathcal{P}_p(X) \to \mathcal{P}_p(X \times X)$ such that $F(\mu, \nu) \in \Pi_o(\mu, \nu)$.

Let $\Lambda \in \mathcal{P}_p(\mathcal{P}_p(X) \times \mathcal{P}_p(X))$ be an optimal coupling of P_1 and P_2, and set $\Omega \coloneqq \mathcal{P}_p(X) \times \mathcal{P}_p(X)$. Let $z_j = (x_j, y_j)$ for $j = 1, 2$. Then,

$$\begin{aligned}
&\int_\Omega W^p_{p,d_{XX}}\big(\delta_{(x_0,y_0)}, F(\mu,\nu)\big)\Lambda(d\mu, d\nu) \\
&\quad= \int_\Omega \left(\int_{(X\times X)^2} d^p_{XX}\big(z_1, z_2\big)\big((\delta_{x_0} \otimes \delta_{y_0}) \otimes F_{\mu,\nu}\big)(dz_1, dz_2) \right) \Lambda(d\mu, d\nu) \\
&\quad= \int_\Omega \Bigg(\int_{X\times X} d^p_X(x_1, x_2)\big(\delta_{x_0} \otimes \mu\big)(dx_1, dx_2) \\
&\qquad\qquad + \int_{X\times X} d^p_X(y_1, y_2)\big(\delta_{y_0} \otimes \nu\big)(dy_1, dy_2) \Bigg) \Lambda(d\mu, d\nu) \\
&\quad= \int_\Omega \left(W^p_{p,d_X}(\delta_{x_0}, \mu) + W^p_{p,d_X}(\delta_{y_0}, \nu) \right) \Lambda(d\mu, d\nu) \\
&\quad< \infty.
\end{aligned}$$

Thus, F is L^p integrable with respect to Λ. Further, $\int_\Omega F(\mu,\nu)\Lambda(d\mu, d\nu)$ is a coupling of $E_X(P_1)$ and $E_X(P_2)$. Indeed,

$$\begin{aligned}
&\left(\int_\Omega F(\mu,\nu)\Lambda(d\mu, d\nu) \right)(A \times X) \\
&\qquad= \int_\Omega \left(\int_{X\times X} \mathbf{1}_{A\times X}(x_1, x_2) F_{\mu,\nu}(dx_1, dx_2) \right) \Lambda(d\mu, d\nu) \\
&\qquad= \int_\Omega F_{\mu,\nu}(A \times X)\Lambda(d\mu, d\nu) \\
&\qquad= \int_\Omega \mu(A)\Lambda(d\mu, d\nu) \\
&\qquad= E_X(P_1)(A).
\end{aligned}$$

Similarly, the second marginal is equal to $E_X(P_2)$. As such,

$$\begin{aligned}
&W^p_{p,d_X}\big(E_X(P_1), E_X(P_2)\big) \\
&\qquad\le \int_{X\times X} d^p_X(x_1, x_2) \left(\int_\Omega F(\mu,\nu)\Lambda(d\mu, d\nu) \right)(dx_1, dx_2) \\
&\qquad= \int_\Omega \left(\int_{X\times X} d^p_X(x_1, x_2) F_{\mu,\nu}(dx_1, dx_2) \right) \Lambda(d\mu, d\nu) \\
&\qquad= \int_\Omega W^p_{p,d_X}(\mu,\nu)\Lambda(d\mu, d\nu) \\
&\qquad= \mathcal{W}^p_p(P_1, P_2),
\end{aligned}$$

which shows that the multiplication, or expectation, map is Lipschitz with Lipschitz constant equal to one. □

It is interesting to note that there does not appear to be a proof of the above lemma using Kantorovich duality, which is the standard method of choice for the case of $p = 1$. Or, at the least, the most immediate and obvious application of Kantorovich duality does not suffice. When $p = 1$, Kantorovich duality is greatly simplified; by the Kantorovich-Rubinstein Theorem (Theorem 1.14 in [59]), we have

$$W^p_{1,d_X}(\mu,\nu) = \sup\left\{\int_X \phi(x)(\mu-\nu)(dx) : \phi \in L^1(|\mu-\nu|), \|\phi\|_{\mathrm{Lip}} \leq 1\right\}$$

where

$$\|\phi\|_{\mathrm{Lip}} := \sup_{x\neq y} \frac{|\phi(x)-\phi(y)|}{d_X(x,y)}.$$

To use Kantorovich duality for general p, we would need to show that for test functions ϕ and ψ satisfying $\phi(x) + \psi(y) \leq d_X(x,y)$, the maps ξ and ζ defined by

$$\xi(\mu) = \int_X \phi(x)\mu(dx) \quad \text{and} \quad \zeta(\nu) = \int_X \psi(x)\nu(dx)$$

satisfy $\xi(\mu) + \zeta(\nu) \leq W^p_{p,d_X}(\mu,\nu)$. The obvious things one might try, however, do not establish this inequality.

Proposition 4.7. *The family of multiplication maps E_X for $X \in$ **Met** defines a natural transformation $E : \mathcal{P}_p^2 \Rightarrow \mathcal{P}_p$.*

Proof. Lemma 4.6 shows that $E_X \in \mathrm{Hom}(\mathcal{P}_p(\mathcal{P}_p(X)), \mathcal{P}_p(X))$. All that remains is to show naturality: Given a morphism $f : X \to Y$, we need

$$E_Y \circ \mathcal{P}_p^2(f) = \mathcal{P}_p(f) \circ E_X.$$

Indeed, letting $P \in \mathcal{P}_p^2(X)$ and $B \in \Sigma_Y$, we have

$$\begin{aligned}
\big(\mathcal{P}_p(f) \circ E_X\big)(P)(B) &= \big(f_\sharp m_X(P)\big)(B) \\
&= E_X(P)\big(f^{-1}(B)\big) \\
&= \int_{\mathcal{P}_p(X)} \mathrm{ev}_{f^{-1}(B)}(\mu)P(d\mu) \\
&= \int_{\mathcal{P}_p(X)} \mathrm{ev}_B(f_\sharp\mu)P(d\mu) \\
&= \int_{\mathcal{P}_p(X)} \big(\mathrm{ev}_B \circ \mathcal{P}_p(f)\big)(\mu)P(d\mu) \\
&= \int_{\mathcal{P}_p(Y)} \mathrm{ev}_B(\nu)\big(\mathcal{P}_p(f)_\sharp P\big)(d\nu) \\
&= \int_{\mathcal{P}_p(Y)} \mathrm{ev}_B(\nu)\big(\mathcal{P}_p^2(f)(P)\big)(d\nu) \\
&= \big(E_Y \circ \mathcal{P}_p^2(f)\big)(P)(B),
\end{aligned}$$

as needed. □

Definition 4.8. Let (X, d_X) be Polish. The **unit map** $\delta_X : X \to \mathcal{P}_p(X)$ is defined by

$$\delta_X(x) = \delta_x.$$

That is, each $x \in X$ is mapped to the Dirac mass δ_x.

Proposition 4.9. *The unit map $\delta : id \Rightarrow \mathcal{P}_p$ is a natural transformation. Additionally, the following diagrams commute:*

$$\begin{tikzcd}
\mathcal{P}_p \arrow[r, Rightarrow, "\delta\mathcal{P}_p"] \arrow[dr, Rightarrow, "id_{\mathcal{P}_p}"'] & \mathcal{P}_p^2 \arrow[d, Rightarrow, "m"] & \mathcal{P}_p \arrow[l, Rightarrow, "\mathcal{P}_p\delta"'] \arrow[dl, Rightarrow, "id_{\mathcal{P}_p}"] \\
& \mathcal{P}_p &
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{P}_p^3 \arrow[r, Rightarrow, "E\mathcal{P}_p"] \arrow[d, Rightarrow, "\mathcal{P}_pE"'] & \mathcal{P}_p^2 \arrow[d, Rightarrow, "E"] \\
\mathcal{P}_p^2 \arrow[r, Rightarrow, "E"'] & \mathcal{P}_p
\end{tikzcd} \tag{7}$$

where the natural transformations $\delta\mathcal{P}_p$ and $\mathcal{P}_p\delta$ are defined componentwise by

$$(\delta\mathcal{P}_p)_X \coloneqq \delta_{\mathcal{P}_p(X)} \quad \textit{and} \quad (\mathcal{P}_p\delta)_X \coloneqq \mathcal{P}_p(\delta_X),$$

and similarly,

$$(E\mathcal{P}_p)_X \coloneqq E_{\mathcal{P}_p(X)} \quad \textit{and} \quad (\mathcal{P}_pE)_X \coloneqq \mathcal{P}_p(E_X).$$

Proof. First, note that δ_X is Lipschitz since $W_{p,d_X}(\delta_{x_1}, \delta_{x_2}) = d_X(x_1, x_2)$. Additionally, given $f : X \to Y$, $x \in X$, and $B \in \Sigma_Y$, the naturality condition holds:

$$\begin{aligned}
\big(\mathcal{P}_p(f) \circ \delta_X\big)(x)(B) &= (f_\sharp \delta_x)(B) \\
&= \delta_x(f^{-1}(B)) \\
&= \delta_{f(x)}(B) \\
&= \big(\delta_Y \circ \mathrm{id}(f)\big)(x)(B).
\end{aligned}$$

Thus, the unit map is a natural transformation. We can similarly verify that the maps $\delta\mathcal{P}_p$, $\mathcal{P}_p\delta$, $E\mathcal{P}_p$, and $\mathcal{P}_pE$ are all natural transformations.

To see that the first diagram commutes, let $\mu \in \mathcal{P}_p(X)$, and let $A \in \Sigma_X$. Then,

$$\begin{aligned}
\big(E \circ \delta\mathcal{P}_p\big)_X(\mu)(A) &= E_X\big(\delta_{\mathcal{P}_p(X)}(\mu)\big)(A) \\
&= \int_{\mathcal{P}_p(X)} \mathrm{ev}_A(\nu)\delta_\mu(d\nu) \\
&= \mu(A) \\
&= \mathrm{id}_{\mathcal{P}_p(X)}(\mu)(A).
\end{aligned}$$

Further,

$$\begin{aligned}
\big(E \circ \mathcal{P}_p\delta\big)_X(\mu)(A) &= E_X\big((\delta_X)_\sharp \mu\big)(A) \\
&= \int_{\mathcal{P}_p(X)} \mathrm{ev}_A(\nu)(\delta_X)_\sharp \mu(d\nu) \\
&= \int_X \big(\mathrm{ev}_A \circ \delta_X\big)(x)\mu(dx)
\end{aligned}$$

$$
\begin{aligned}
&= \int_X \delta_x(A)\mu(dx) \\
&= \mu(A) \\
&= \mathrm{id}_{\mathcal{P}_p(X)}(\mu)(A),
\end{aligned}
$$

as needed.

To see that the second diagram commutes, let $\Lambda \in \mathcal{P}_p^3(X)$, and let $A \in \Sigma_X$. Then,

$$
\begin{aligned}
\big(E \circ E\mathcal{P}_p\big)_X(\Lambda)(A) &= E_X\big(E_{\mathcal{P}_p(X)}(\Lambda)\big)(A) \\
&= \int_{\mathcal{P}_p(X)} \mathrm{ev}_A(\mu) E_{\mathcal{P}_p(X)}(\Lambda)(d\mu) \\
&= \int_{\mathcal{P}_p^2(X)} \left(\int_{\mathcal{P}_p(X)} \mathrm{ev}_A(\mu) P(d\mu) \right) \Lambda(dP) \\
&= \int_{\mathcal{P}_p^2(X)} E_X(P)(A)\Lambda(dP) \\
&= \int_{\mathcal{P}_p^2(X)} (\mathrm{ev}_A \circ E_X)(P)\Lambda(dP) \\
&= \int_{\mathcal{P}_p(X)} \mathrm{ev}_A(\mu)(E_X)_\sharp \Lambda(d\mu) \\
&= \big(E_X \circ \mathcal{P}_p(E_X)\big)(\Lambda)(A) \\
&= \big(E \circ \mathcal{P}_p E\big)_X(\Lambda)(A).
\end{aligned}
$$

□

Theorem 4.10. *$(\mathcal{P}_p, \delta, m)$ is a monad on the category* **Met**.

Proof. This is the content of Propositions 4.4, 4.7, and 4.9. □

In [28], Fritz and Perrone showed that the 1-Kantorovich monad actually has the structure of a symmetric bimonoidal monad with monoidal structure given by the joint product and with opmonoidal structure given by marginalization. It is likely that this additional structure applies to p-Kantorovich monads for general $p \geq 1$, but we leave this question for future work.

4.1. **The Kleisli Category.** The Kleisli category can be defined for any monad, though here we will only state the construction for $\mathcal{P}_p$. The Kleisli category $\mathbf{Kl}(\mathcal{P}_p)$ of $\mathcal{P}_p$ is defined as follows: The objects of $\mathbf{Kl}(\mathcal{P}_p)$ are the same as the objects in $\mathbf{Met}_p$, and the set of morphisms $\mathbf{Kl}(\mathcal{P}_p)(X, Y)$ is the same as $\mathrm{Hom}(X, \mathcal{P}_p(Y))$. That is, a morphism in the Kleisli category is a Lipschitz map $F : X \to \mathcal{P}_p(Y)$. Further, the Kleisli composition $G \odot F$ of $F : X \to \mathcal{P}_p(Y)$ and $G : Y \to \mathcal{P}_p(Z)$ is given by

$$
\begin{aligned}
(G \odot F)(x)(E) &= \big(E_Z \circ \mathcal{P}_p(G) \circ F\big)(x)(E) \\
&= \int_{\mathcal{P}_p(Y)} \mathrm{ev}_E(\mu)(G_\sharp F_x)(d\mu)
\end{aligned}
$$

$$= \int_Y \big(\mathrm{ev}_E \circ G\big)(y) F_x(dy)$$
$$= \int_Y G(E|y) F(dy|x).$$

That is, we use a change of variables to define the composition as an average over Y. We use this notation in Section 5 in the following context: We can view a probability measure $\mu \in \mathcal{P}_p(X)$ as an arrow in the Kleisli category $\mu : I \to \mathcal{P}_p(X)$. Then, if $\int_X \big(\delta_x \otimes F_x\big)\mu(dx) = \gamma$, we have $F \odot \mu = \gamma^2$.

To conclude this section, we note that while the monadic framework developed here is sufficient for capturing the categorical content of, for example, the iterated Wasserstein space construction, it nonetheless falls short of providing a comprehensive structural framework for Bayesian inference. In particular, as will be seen in Section 5, the Bayesian inverse map depends on the prior. As such, more elaborate structures like lenses, for example, are needed to model this dependence precisely in a categorical framework. See [11], [50], and [51] for more on this. Note, though, that our results in Section 5 are independent of these considerations.

5. Bayesian Inference

In this section, we define Bayesian inverse maps using Markov kernels and the Disintegration Theorem. Our main results are Theorem 5.4, Proposition 5.7, and Theorem 5.10. In Theorem 5.4, we give a simplified proof of a generalized Bayes's Law, while in Proposition 5.7 and Theorem 5.10, we show that, under certain conditions, the expected posterior distribution converges under repeated updates to a fixed point of a certain map on the space of posterior distributions.

To begin, recall that for $F : X \to \mathcal{P}_p(Y)$ and $\mu \in \mathcal{P}_p(X)$, we write $F \odot \mu$ for the probability measure in $\mathcal{P}_p(Y)$ defined by the Kleilsi composition of $\mu : I \to \mathcal{P}_p(X)$ with F. More concretely, for $B \in \Sigma_Y$, we have

$$(F \odot \mu)(B) = \int_X F(B|x)\mu(dx).$$

We define the **Kleisli image** of F to be

$$F \odot \mathcal{P}_p(X) \coloneqq \{F \odot \mu : \mu \in \mathcal{P}_p(X)\} \subseteq \mathcal{P}_p(Y).$$

Also, note that if $\gamma = \int_X \big(\delta_x \otimes F_x\big)\mu(dx)$, then $\gamma^2 = F \odot \mu$, where γ^2 is the second marginal.

Definition 5.1. Let $F \in L^p_\mu(X, \mathcal{P}_p(Y))$, and set $\nu \coloneqq F \odot \mu$. Then, the **Bayesian inverse** $F^\dagger_\mu$ of F with respect to μ is the ν-almost surely unique Markov kernel in $L^p_\nu(Y, \mathcal{P}_p(X))$ satisfying

$$\int_X \big(\delta_x \otimes F(x)\big)\mu(dx) = \int_Y \big(F^\dagger_\mu(y) \otimes \delta_y\big)(F \odot \mu)(dy). \tag{8}$$

$F^\dagger_\mu$ is guaranteed to exist by Theorem 3.16.

Thus, we see that a Bayesian inverse map is a special case of a disintegration. It is important to note, though, that the Bayesian inverse $F^\dagger_\mu$ depends on the prior μ. At least when F is Lipschitz, we can guarantee that this dependence is measurable. To show this, we appeal to the work of Bogachev and Malofeev, who study parametric dependence of Kantorovich problems and disintegrations in [10]. Their work is highly technical; we therefore merely employ one of their results and refer the reader to [10] for further details. First, we need the following proposition.

Proposition 5.2. *Let $F : X \to \mathcal{P}_p(Y)$ be Lipschitz. Define maps I_F and $I_{F^\dagger}$ from $\mathcal{P}_p(X)$ to $\mathcal{P}_p(X \times Y)$ by*

$$I_F(\mu) \coloneqq \int_X \big(\delta_x \otimes F(x)\big)\mu(dx),$$

and

$$I_{F^\dagger}(\mu) \coloneqq \int_Y \big(F^\dagger_\mu(y) \otimes \delta_y\big)(F \odot \mu)(dy).$$

Then, I_F and $I_{F^\dagger}$ are both Lipschitz with respect to W_{p,d_X} and $W_{p,d_{XY}}$.

Proof. Since F is Lipschitz, it is L^p integrable for any $\mu \in \mathcal{P}_p(X)$, and thus, I_F is well-defined. Let K be the Lipschitz constant for F. Using the Disintegration Theorem together with Lemma 3.18, we have

$$\begin{aligned}
&W^p_{p,d_{XY}}\big(I_{F^\dagger}(\mu_1), I_{F^\dagger}(\mu_2)\big) \\
&\quad = W^p_{p,d_{XY}}\big(I_F(\mu_1), I_F(\mu_2)\big) \\
&\quad \le \inf_{\pi:\Pi(\mu_1,\mu_2)} \int_{X^2} \Big(d^p_X(x_1,x_2) + W^p_{p,d_Y}\big(F(x_1), F(x_2)\big)\Big)\,\pi(dx_1, dx_2) \\
&\quad \le \inf_{\pi:\Pi(\mu_1,\mu_2)} \int_{X^2} \big(d^p_X(x_1,x_2) + K^p d^p_X(x_1,x_2)\big)\,\pi(dx_1, dx_2) \\
&\quad = (1+K^p) W^p_{p,d_X}(\mu_1,\mu_2),
\end{aligned}$$

as needed.

□

Proposition 5.3. *Let $F : X \to \mathcal{P}_p(Y)$ be Lipschitz. Define $F^\dagger : \mathcal{P}_p(X) \times Y \to \mathcal{P}_p(X)$ by*

$$F^\dagger(\mu, y) \coloneqq F^\dagger_\mu(y).$$

Then, $F^\dagger$ is Borel measurable with respect to $\Sigma_{\mathcal{P}_p(X)} \times \Sigma_Y$ and $\Sigma_{\mathcal{P}_p(X)}$, where $\Sigma_{\mathcal{P}_p(X)}$ is the Borel σ-algebra on $\mathcal{P}_p(X)$ induced by the Wasserstein topology.

Proof. This follows from Corollary 3.6 in [10] together with Proposition 5.2 above. □

Naturally, it would be worth trying to relax the conditions needed for measurable dependence. Additionally, we might ask when this dependence is actually stronger. When, for example, is $F^\dagger : \mathcal{P}_p(X) \times Y \to \mathcal{P}_p(X)$ (locally) Lipschitz with respect to the product metric $W_{p,d_X} \otimes d_Y$? As noted in Section 4, it is precisely this dependence on the prior that greatly complicates the categorical picture of Bayesian inference,

and we again refer to such treatments as [11], [50], or [51] for more on how to move beyond the limitations of a purely monadic framework to capture the full categorical content of Bayesian inference.

Now, while (8) defines $F_\mu^\dagger$ via an equality of two joint distributions, in practice we typically compute posterior distributions by applying Bayes's Law, or some computational approximation thereof. Theorem 5.4 shows that a generalized Bayes's Law does indeed hold in our more abstract setting. In [31], Jost et al. prove a generalized Bayes's Law of the same form as the one below under more general assumptions. Here, though, we give a much simpler proof under more restrictive assumptions using our approximation machinery developed in Section 3. Specifically, we require $F : X \to \mathcal{P}_p(Y)$ to be continuous so that the following analog of the Lebesgue Differentiation Theorem holds: For any $x_0 \in \mathrm{supp}(\mu)$ and any $E \in \Sigma_Y$,

$$F(x_0)(E) = \lim_{s\to 0} \frac{1}{\mu\big(B_s(x_0)\big)} \int_{B_s(x_0)} F_x(E)\mu(dx).$$

This equality likely holds under integrability alone, but proving this would require a suite of additional foundational results beyond the scope of the present work.

In the theorem below, and throughout the rest of the work, we will omit the copy map when writing $F^{\otimes n} \circ \mathrm{copy}_X^n : X \to \mathcal{P}_p(Y^n)$ and instead write $F^{\otimes n} : X \to \mathcal{P}_p(Y^n)$. Additionally, we denote the Bayesian inverse of this map by $(F_{\mu_0}^{\otimes n})^\dagger : Y^n \to \mathcal{P}_p(X)$, and we use $B_s(-)$ to denote an open ball of radius s with respect to the product metric in either X^n or Y^n for $n \geq 1$, depending on the context.

Theorem 5.4 (**Bayes's Law**). *Let $F : X \to \mathcal{P}_p(Y)$ be continuous and L^p-integrable with respect to some prior distribution $\mu_0 \in \mathcal{P}_p(X)$. Let $y_1, \ldots, y_n \in Y$ be arbitrary data points in the support of $F \odot \mu_0$, not necessarily distinct. Then, the posterior distribution given by $\mu_n \coloneqq F_{\mu_{n-1}}^\dagger(y_n)$ is well-defined, and, in fact,*

$$\mu_n = F_{\mu_{n-1}}^\dagger(y_n) = (F_{\mu_0}^{\otimes n})^\dagger(y_1, \ldots, y_n).$$

Moreover, we have the following generalized Bayes's Law: For any $E \in \Sigma_X$,

$$\mu_n(E) = \lim_{s\to 0} \frac{\int_E \prod_{k=1}^n F_x\big(B_s(y_k)\big)\mu_0(dx)}{\int_X \prod_{k=1}^n F_x\big(B_s(y_1)\big)\mu_0(dx)}. \tag{9}$$

Proof. Let F, μ_0, and $y_1, \ldots, y_n$ satisfy the stated assumptions. Note that in practice, each y_j is assumed to be sampled independently from some $\tau \in \mathcal{P}_p(Y)$ with $\mathrm{supp}(\tau) \subseteq \mathrm{supp}(F \odot \mu_0)$. However, we need not introduce τ explicitly to establish the result. First, we'll show that the generalized Bayes's Law holds for measures of the form $(F_{\mu_0}^{\otimes n})^\dagger(y_1, \ldots, y_n)$. Let $\gamma_0 \in \mathcal{P}_p(X \times Y)$ denote the joint distribution defined by

$$\gamma_0(E \times B) \coloneqq \int_E F(B|x)\mu_0(dx).$$

Given that $y_1 \in \operatorname{supp}(F \odot \mu_0)$, we use the Disintegration Theorem (3.16) together with Lebesgue-type differentiation to write

$$\begin{aligned}\mu_1(E) = F^\dagger_{\mu_0}(y_1)(E) &= \lim_{s\to 0} \frac{\int_{B_s(y_1)} F^\dagger_{\mu_0}(E|y)(F\odot\mu_0)(dy)}{(F\odot\mu_0)\big(B_s(y_1)\big)} \\ &= \lim_{s\to 0} \frac{\int_{B_s(y_1)} F^\dagger_{\mu_0}(E|y)(F\odot\mu_0)(dy)}{\int_{B_s(y)} F^\dagger_{\mu_0}(X|y)(F\odot\mu_0)(dy)} \\ &= \lim_{s\to 0} \frac{\gamma_0\big(E\times B_s(y_1)\big)}{\gamma_0\big(X\times B_s(y_1)\big)} \\ &= \lim_{s\to 0} \frac{\int_E F_x\big(B_s(y_1)\big)\mu_0(dx)}{\int_X F_x\big(B_s(y_1)\big)\mu_0(dx)}.\end{aligned} \tag{10}$$

As noted in Remark 3.14, $F^{\otimes n} : X \to \mathcal{P}_p(Y^n)$ is L^p-integrable with respect to μ_0. Also, since we are assuming that $y_j \in \operatorname{supp}(F\odot\mu_0)$ for $j \in \{1,\dots,n\}$, it follows that $(y_1,\dots,y_n) \in \operatorname{supp}(F^{\otimes n}\odot\mu_0)$. Thus, the same computation from above shows that

$$\begin{aligned}(F^{\otimes n}_{\mu_0})^\dagger(y_1,\dots,y_n)(E) &= \lim_{s\to 0} \frac{\int_E F^{\otimes n}_x\big(B_s(y_1,\dots,y_n)\big)\mu_0(dx)}{\int_X F^{\otimes n}_x\big(B_s(y_1,\dots,y_n)\big)\mu_0(dx)} \\ &= \lim_{s\to 0} \frac{\int_E F^{\otimes n}_x\big(B_s(y_1),\dots,B_s(y_n)\big)\mu_0(dx)}{\int_X F^{\otimes n}_x\big(B_s(y_1),\dots,B_s(y_n)\big)\mu_0(dx)} \\ &= \lim_{s\to 0} \frac{\int_E \prod_{k=1}^n F_x\big(B_s(y_k)\big)\mu_0(dx)}{\int_X \prod_{k=1}^n F_x\big(B_s(y_1)\big)\mu_0(dx)}.\end{aligned} \tag{11}$$

Next, we'll show that $F^\dagger_{\mu_1}(y_2)$ is well-defined and that it is equal to $(F^{\otimes}_{\mu_0})(y_1,y_2)$. The general case will then follow from induction. First, in general, $G(y)$ is absolutely continuous with respect to $G\odot\nu$ for any Markov kernel $G \in L^p_\nu(Y,\mathcal{P}_p(X))$. As such, $\mu_1 = F^\dagger_{\mu_0}(y_1)$ is absolutely continuous with respect to $F^\dagger_{\mu_0}\odot(F\odot\mu_0) = \mu_0$, and F is integrable with respect to μ_1, since it is essentially bounded. Hence, $F^\dagger_{\mu_1}(y_2)$ is well-defined. Now, let $\{F_k\}_{k=1}^\infty$ be a sequence of countable simple Markov kernels converging uniformly in the Wasserstein metric to F, as constructed in Theorem 3.9. That is, F_k has the form $F_k = \sum_{m=1}^\infty \nu^k_m \mathbf{1}_{A^k_m}$, where $\nu^k_m \in \mathcal{P}_p(Y)$ and $\{A^k_m\}_{m=1}^\infty$ is a pairwise disjoint partition of X. Set

$$Z^k_1 := \int_X F_k\big(B_s(y_1)|x\big)\mu_0(dx) = \sum_{m=1}^\infty \nu^k_m(B_s(y_1))\mu_0(A^k_m).$$

Also, set $\mu^k_1 := F^\dagger_{k,\mu_0}(y_1)$. Then, for any k and any $E \in \Sigma_X$, we apply (10) to write

$$F^\dagger_{k,\mu_1}(y_2)(E) = \lim_{s\to 0} \frac{\int_E F_k\big(B_s(y_2)|x\big)\mu^k_1(dx)}{\int_X F_k\big(B_s(y_2)|x\big)\mu^k_1(dx)}$$

$$
\begin{aligned}
&= \lim_{s\to 0} \sum_{j=1}^{\infty} \frac{\nu_j^k\big(B_s(y_2)\big)\mu_1^k(E\cap A_j^k)}{\sum_{i=1}^{\infty}\nu_i^k\big(B_s(y_2)\big)\mu_1^k(A_i^k)} \\
&= \lim_{s\to 0} \sum_{j=1}^{\infty} \frac{\nu_j^k\big(B_s(y_2)\big)\Big[\sum_{\ell=1}^{\infty}(1/Z_1^k)\nu_\ell^k(B_s(y_1))\mu_0(E\cap A_j^k\cap A_\ell^k)\Big]}{\sum_{i=1}^{\infty}\nu_i^k\big(B_s(y_2)\big)\Big[\sum_{\ell=1}^{\infty}(1/Z_1^k)\nu_\ell^k(B_s(y_1))\mu_0(A_i^k\cap A_\ell^k)\Big]} \\
&= \lim_{s\to 0} \sum_{j=1}^{\infty} \frac{\nu_j^k\big(B_s(y_2)\big)\nu_j^k\big(B_s(y_1)\big)\mu_0(E\cap A_j^k)}{\sum_{i=1}^{\infty}\nu_i^k\big(B_s(y_2)\big)\nu_i^k\big(B_s(y_1)\big)\mu_0(A_j^k)} \\
&= \lim_{s\to 0} \frac{\int_E F_k^{\otimes 2}\big(B_s(y_1),B_s(y_2)|x\big)\mu_0(dx)}{\int_X F_k^{\otimes 2}\big(B_s(y_1),B_s(y_2)|x\big)\mu_0(dx)} \\
&= (F_{k,\mu_0}^{\otimes 2})^{\dagger}(y_1,y_2)(E).
\end{aligned}
$$

By induction, a similar computation to the above shows that

$$F_{k,\mu_{n-1}}^{\dagger}(y_n) = (F_{k,\mu_0}^{\otimes n})^{\dagger}(y_1,\dots,y_n)$$

for any k and any n. Then, since $\{F_k\}_{k=1}^{\infty}$ converges to F uniformly, and since this convergence is uniform with respect to s, we can pass the limit with respect to k through the integral and then interchange the limits with respect to s and k. Thus,

$$\mu_n = F_{\mu_{n-1}}^{\dagger}(y_n) = (F_{\mu_0}^{\otimes n})^{\dagger}(y_1,\dots,y_n),$$

and by (11), it follows that

$$\mu_n(E) = \lim_{s\to 0} \frac{\int_E \prod_{k=1}^{n} F_x\big(B_s(y_k)\big)\mu_0(dx)}{\int_X \prod_{k=1}^{n} F_x\big(B_s(y_1)\big)\mu_0(dx)}.$$

□

Next we study convergence of the posterior distribution under repeated Bayesian updates. More specifically, we will work with the expected update, rather than the exact update.

Definition 5.5. Let $F : X \to \mathcal{P}_p(Y)$, and let $\mu_0 \in \mathcal{P}_p(X)$ and $\tau \in \mathcal{P}_p(Y)$ be arbitrary. Then, the **expected Bayesian update** of F with respect to μ_0 and τ is

$$\bar{\mu}_1 = F_{\mu_0}^{\dagger} \odot \tau, \quad \text{and} \quad \bar{\mu}_n \coloneqq F_{\bar{\mu}_{n-1}}^{\dagger} \odot \tau$$

for $n \geq 2$.

Proving that repeated Bayesian updates, whether expected or exact, actually converge is greatly complicated by the dependence of the inverse map on the prior. Indeed, naive approaches, in our experience, invariably seem to run into circularity, wherein proving convergence of the iteratively updated posterior can be reduced to any number of statements whose truth ultimately depends on exactly the conjectured convergence of the posteriors. We therefore consider two approaches for breaking out of this loop.

The first approach, as developed in Proposition 5.7, is conceptually elegant, as it reduces convergence to an application of the Banach Fixed Point Theorem. However,

the assumptions are fairly strict, and more work is needed to determine when they hold in practice. The second approach, as developed in Theorem 5.10, gives more explicit conditions for convergence, but the proof is rather labor-intensive. To set up both of these approaches, we frame Bayesian inference as an abstract Markov process as follows.

Definition 5.6. Let $F : X \to \mathcal{P}_p(Y)$ be a Lipschitz Markov kernel, and let $\tau \in \mathcal{P}_p(Y)$. The **(Bayesian) inference process** of F with respect to τ is the map $F_\tau : \mathcal{P}_p(X) \to \mathcal{P}_p^2(X)$ defined by

$$F_\tau(\mu) = (F_\mu^\dagger)_\sharp \tau.$$

Note that since F is Lipschitz, it is integrable with respect to any $\mu \in \mathcal{P}_p(X)$. Thus, F_τ is well-defined, since $F_\mu^\dagger$ always exists.

Given a prior μ, $F_\tau(\mu)$ defines a distribution of distributions in the iterated Wasserstein space $\mathcal{P}_p^2(X)$. Sampling from this distribution gives us a posterior distribution, and, in particular, if we take the expectation $E_X : \mathcal{P}^2(X) \to \mathcal{P}_p(X)$, we arrive at the expected posterior distribution: $(E_X \circ F_\tau)(\mu) = E_X \circ (F_\mu^\dagger)_\sharp \tau = F_\mu^\dagger \odot \tau$. Moreover, this update only depends on the current state. As such, we can view Bayesian inference as an abstract Markov chain on $\mathcal{P}_p(X)$.

We do not formalize this perspective fully here, but it is, we think, worth pursuing more carefully in future work. In particular, we could then relate convergence properties of the update process to synthetic Ricci curvature by appealing to the general theory of synthetic Ricci curvature for Markov chains in metric measure spaces, as first developed by Ollivier in [41]. Proposition 5.7 below ensures convergence under the condition that $E_X \circ F_\tau$ is a contraction map, which, in turn, implies that the synthetic Ricci curvature of F_τ, defined by

$$\kappa(\mu_1, \mu_2) = 1 - \frac{\mathcal{W}_X\big(F_\tau(\mu_1), F_\tau(\mu_2)\big)}{W_{p,d_X}(\mu_1, \mu_2)},$$

is non-negative, where $\mathcal{W}_X$ is the iterated Wasserstein metric. Fully exploring this connection is, as noted, left for future work.

Proposition 5.7. *Let $F : X \to \mathcal{P}_p(Y)$ be Lipschitz, and let $\tau \in F \odot \mathcal{P}_p(X)$. Then, if $F_\tau : \mathcal{P}_p(X) \to \mathcal{P}_p^2(X)$ is Lipschitz with Lipschitz constant $0 < K < 1$, the expected Bayesian update converges to a unique optimal value starting from any prior.*

Proof. Let $F : X \to \mathcal{P}_p(Y)$ be Lipschitz with Lipschitz constant L, and let $\tau \in F \odot \mathcal{P}_p(X)$. By Lemma 4.6, $E_X : \mathcal{P}_p^2(X) \to \mathcal{P}_p(X)$ is Lipschitz with Lipschitz constant equal to one. As such, $E_X \circ F_\tau : P_p(X) \to \mathcal{P}_p(X)$ is a contraction map, since we are assuming F_τ is Lipschitz with constant $0 < K < 1$. Then, by the Banach Fixed-Point Theorem, $E_X \circ F_\tau$ has a unique fixed point $\mu_* \in \mathcal{P}_p(X)$. That is, $(E_X \circ F_\tau)(\mu_*) = F_{\mu_*}^\dagger \odot \tau = \mu_*$.

Now, let $\mu \in \mathcal{P}_p(X)$ be arbitrary, and let $E \in \Sigma_X$. Then,

$$\big(F_\mu^\dagger \odot (F \odot \mu)\big)(E) = \int_Y F_\mu^\dagger(E|y)(F \odot \mu)(dx)$$

$$
\begin{aligned}
&= \left(\int_Y \big(F_\mu^\dagger(y) \otimes \delta_y\big)(F \odot \mu)(dx) \right)(E \times Y) \\
&= \left(\int_X \big(\delta_x \otimes F(x)\big)\mu(dx) \right)(E \times Y) \\
&= \int_E F(Y|x)\mu(dx) \\
&= \mu(E).
\end{aligned}
$$

Thus, $F_\mu^\dagger \odot (F \odot \mu) = \mu$ for any $\mu \in \mathcal{P}_p(Y)$. As such, given $\mu \in \mathcal{P}_p(X)$ with $F \odot \mu = \tau$, we have

$$\mu = F_\mu^\dagger \odot (F \odot \mu) = F_\mu^\dagger \odot \tau = (E_X \circ F_\tau)(\mu). \tag{12}$$

That is, μ is a fixed-point of $E_X \circ F_\tau$, and by uniqueness of fixed points, $\mu = \mu_*$. Hence, given any prior μ_0,

$$\lim_{n\to\infty} \bar{\mu}_n = \lim_{n\to\infty} (E_X \circ F_\tau)(\bar{\mu}_{n-1}) = \mu_*,$$

and $F \odot \mu_* = F \odot \mu = \tau$. □

Of course, the question remains: When does F_τ satisfy the needed Lipschitz condition? In our second approach to convergence, we take a hands-on approach and show that, under certain conditions, the expected Bayesian updated converges to a fixed point of the map $E_X \circ F_\tau$. From (12), we know that $F \odot \tau = \mu$ always implies $F_\mu^\dagger \odot \tau = \mu$ for any μ and τ. Thus, in situations where $F \odot \mu = \tau$ if and only if $F_\mu^\dagger \odot \tau = \mu$, Theorem 5.10 implies that the expected update exactly recovers the true distribution.

Finding sufficient conditions for $F_\mu^\dagger \odot \tau = \mu$ to imply that $F \odot \mu = \tau$ is left as an open problem. Alternatively, we might ask when the equality $F_\mu^\dagger \odot \tau = \mu$ implies that

$$\mu \in \arg\min_\sigma W_{p,d_Y}(F \odot \sigma, \tau).$$

We again leave this as an open problem. Independently of these considerations, we will show that the expected Bayesian update converges to a fixed point under the following regularity conditions.

Definition 5.8. Let $F : X \to \mathcal{P}_p(Y)$ be a Markov kernel. The **evaluation map** associated to F is the map $\mathrm{ev}_F : X \times [0,1] \times Y \to [0,1]$ defined by

$$\mathrm{ev}_F(x, s, y) = F_x\big(B_s(y)\big).$$

Further, we call $\mathrm{ev}_F(x, \cdot, y) : [0,1] \to [0,1]$ the **contraction map** of F at $x \in X$ and $y \in Y$. We say that F has **continuously varying bi-Lipschitz** contraction maps if for all $x \in X$ and $y \in Y$, the contraction map $\mathrm{ev}_F(x, \cdot, y)$ is bi-Lipschitz and differentiable at $s = 0$ with $\frac{\partial}{\partial s}_{|s=0} \mathrm{ev}_F(x, s, y)$ jointly continuous in x and y.

Note that, by continuity, for any compact sets $K_1 \subseteq X$ and $K_2 \subseteq Y$, the quantities

$$L^0_{K_1,K_2} \coloneqq \min\{L^0_{x,y} : x \in K_1, y \in K_2\}, \text{ and}$$

$$L^1_{K_1,K_2} \coloneqq \max\{L^1_{x,y} : x \in K_1, y \in K_2\}$$

are well-defined, where $L^0_{x,y}$ and $L^1_{x,y}$ are, respectively, the lower and upper Lipschitz constants for $\mathrm{ev}_F(x,\cdot,y)$.

Definition 5.9. We say that a Markov kernel $F : X \to \mathcal{P}_p(Y)$ is **regular** if the following conditions hold:

(1) The evaluation map ev_F is jointly continuous with respect to the product metric $\left(d_X^p + |\cdot|^p + d_Y^p\right)^{1/p}$ on $X \times [0,1] \times Y$.
(2) F has continuously varying bi-Lipschitz contraction maps.

Theorem 5.10. *Let $F : X \to \mathcal{P}_p(Y)$ be a Lipschitz Markov kernel satisfying the regularity conditions in definition (5.9). Fix some true distribution $\tau \in \mathcal{P}_p(Y)$ with compact support. Further, let $\mu_0 \in \mathcal{P}_p(X)$ be any prior distribution with compact support such that $supp(\tau) \subseteq supp(F_x)$ for all $x \in supp(\mu_0)$. Then, $\bar{\mu} \coloneqq \lim_{n\to\infty} \bar{\mu}_n$ is a well-defined measure in $\mathcal{P}_p(X)$ with $F^\dagger_{\bar{\mu}} \odot \tau = \bar{\mu}$. That is, the expected Bayesian update converges to a fixed point of the map $E_X \circ F_\tau$.*

Proof. Set $\bar{\mu}_1 \coloneqq F^\dagger_{\mu_0} \odot \tau$, and set $\bar{\mu}_n \coloneqq F^\dagger_{\bar{\mu}_{n-1}} \odot \tau$ for $n \geq 2$. We'll show that $\bar{\mu}(E) \coloneqq \lim_{n\to\infty} \bar{\mu}_n(E)$ is a well-defined measure in $\mathcal{P}_p(X)$ with $(E_X \circ F_\tau)(\bar{\mu}) = \bar{\mu}$. More specifically, we'll rewrite the expected update $\bar{\mu}_n$ using Bayes's Law and show that the resulting integral over Y^n with respect to the product measure $\tau^{\otimes n}$ converges.

In applying Bayes's Law, we will need to account for the limit with respect to the parameter s. First, though, we work with some fixed $s > 0$, and we indicate this dependence below with the notation μ_n^s. Using the generalized Bayes's Law of Theorem 5.4, we have

$$\mathbb{E}_{\tau^{\otimes n}}[\mu_n^s(E)] = \int_{Y^n} \frac{\int_E \prod_{k=1}^n F_x\big(B_s(y_k)\big)\mu_0(dx)}{\int_X \prod_{k=1}^n F_x\big(B_s(y_k)\big)\mu_0(dx)} \tau^{\otimes n}(d\mathbf{y}_n). \tag{13}$$

Our strategy will be to decompose this integral using nested partitions on which we can give nice upper and lower approximations. As the partitions are refined, we can squeeze (13) between these two bounding, convergent approximations. Let $\mathcal{A} = \{\mathcal{A}_N\}_{N=1}^\infty$ and $\mathcal{C} = \{\mathcal{C}_M\}_{M=1}^\infty$ be nested partitions on $\mathrm{supp}(\mu_0)$ and $\mathrm{supp}(\tau)$, respectively, given by Lemma 3.7. Since $\mathrm{supp}(\tau)$ is compact, $|\mathcal{C}_M|$ is finite for all M. Let $N = M$ be fixed, and set $E_J \coloneqq E \cap A_J$. Choose some enumeration $K_1, \ldots, K_r$ of the multi-indices in $\mathcal{C}_N$ where $r = |\mathcal{C}_N|$. For simplicity, we will write C_{K_j} as C_j. Further, let $[r]$ denote the set $\{1, \ldots, r\}$, and set

$$\mathcal{L}_k^r \coloneqq \{L = (\ell_1, \ldots, \ell_k) \in [r]^k : \ell_i \neq \ell_j \text{ for } i \neq j\}.$$

$\mathcal{L}_k^r$ serves as our index set[1] for specifying all $r!/(r-k)!$ arrangements of k distinct sets chosen from the r sets of $\mathcal{C}_N$. We will also use the notation $T = (t_1, \ldots, t_k)$ for

[1]Since the integrand is permutation invariant, it is also possible to sum over strictly increasing sequences $\ell_1 < \ell_2 < \cdots < \ell_k$ by multiplying the resulting integral by an appropriate multinomial coefficient. This, however, complicates the asymptotic analysis. It is simpler, for example, to treat the integral over $C_1 \times C_2$ as distinct from the integral over $C_2 \times C_1$.

specifying a vector of multiplicities, and we write $\hat{T} := \sum_{j=1}^{k} t_j$, and $\hat{T}(\ell) := \sum_{j\neq\ell} t_j$. Additionally, for compactness, we write

$$f_n^s(x|\mathbf{y}_n) := \prod_{k=1}^{n} F_x\big(B_s(y_k)\big),$$

and we set

$$I_n^s(\mathbf{y}_n|E) := \frac{\int_E f_n^s(x|\mathbf{y}_n)\mu_0(dx)}{\int_X f_n^s(x|\mathbf{y}_n)\mu_0(dx)} \quad \text{and} \quad I_{N,n}^s(\mathbf{y}_n|E) := \frac{\int_E f_n^s(x|\mathbf{y}_n)\mu_0(dx)}{\sum_{I:\mathcal{A}_N}\int_{A_I} f_n^s(x|\mathbf{y}_n)\mu_0(dx)}.$$

Now, for any n, note that

$$\begin{aligned}\mathcal{C}_N^n &:= \big\{C_{\ell_1}^{t_1}\times\cdots\times C_{\ell_k}^{t_k} : \hat{T}=n, L\in\bigcup_{k=1}^{r}\mathcal{L}_k^r\big\}\\ &= \bigcup_{k=1}^{r}\big\{C_{\ell_1}^{t_1}\times\cdots\times C_{\ell_k}^{t_k} : \hat{T}=n, L\in\mathcal{L}_k^r\big\}\end{aligned}$$

is a partition of Y^n. In what follows, we decompose (13) by first splitting the inner-integrals in both the numerator and denominator using $\mathcal{A}_N$. Then, the integral over Y^n is decomposed according to the number of distinct factors $C_{\ell_1},\dots,C_{\ell_k}$ chosen from $\mathcal{C}_N$, and finally, the vector $T=(t_1,\dots,t_k)$ of multiplicities is chosen from $\mathbb{Z}_+^k$ under the constraint that $\hat{T}=n$. In order to analyze the limit of (13) as $n\to\infty$, it will be more convenient to fix the first $k-1$ coordinates of T and write $t_k = n-(t_1+\cdots+t_{k-1}) = n-\hat{T}(k)$. This corresponds to traversing $\mathbb{Z}_+^k$ along one basis direction at a time, and we let D_n^k denote the event that $\hat{T}(k)\le n-1$. Putting all this together, we have

$$\begin{aligned}&\mathbb{E}_{\tau^{\otimes n}}[\mu_n^s(E)] \qquad (14)\\ &= \sum_{J:\mathcal{A}_N}\int_{Y^n}\frac{\int_{E_J} f_n^s(x|\mathbf{y}_n)\mu_0(dx)}{\int_X f_n^s(x|\mathbf{y}_n)\mu_0(dx)}\tau^{\otimes n}(d\mathbf{y}_n)\\ &= \sum_{J:\mathcal{A}_N}\int_{Y^n}\frac{\int_{E_J} f_n^s(x|\mathbf{y}_n)\mu_0(dx)}{\sum_{I:\mathcal{A}_N}\int_{A_I} f_n^s(x|\mathbf{y}_n)\mu_0(dx)}\tau^{\otimes n}(d\mathbf{y}_n)\\ &= \sum_{J:\mathcal{A}_N}\sum_{k:[r]}\sum_{L:\mathcal{L}_k^r}\sum_{T:\mathbb{Z}_+^k}\mathbf{1}_{\{\hat{T}=n\}}\int_{C_{\ell_1}^{t_1}\times\cdots\times C_{\ell_k}^{t_k}} I_{N,n}^s(\mathbf{y}_n|E_J)\tau^{\otimes n}(d\mathbf{y}_n)\\ &= \sum_{J:\mathcal{A}_N}\sum_{k:[r]}\sum_{L:\mathcal{L}_k^r}\sum_{t_1:\mathbb{Z}_+}\cdots\sum_{t_{k-1}:\mathbb{Z}_+}\mathbf{1}_{\{D_n^k\}}\int_{C_{\ell_1}^{t_1}\times\cdots\times C_{\ell_k}^{n-\hat{T}(k)}} I_{N,n}^s(\mathbf{y}_n|E_J)\tau^{\otimes n}(d\mathbf{y}_n).\end{aligned}$$

Next we build upper and lower approximations for the above decomposition. Set

$$H_{E,n}^s = H_E^s(\mathbf{y}_n) := \int_E f_n^s(x|\mathbf{y}_n)\mu_0(dx).$$

When $E = A_I$, we abbreviate $H^s_{A_I,n}$ as $H^s_{I,n}$. Then, for any n,

$$I^s_{N,n}(\mathbf{y}_n|E_J) = \frac{\int_{E_J} f^s_n(x|\mathbf{y}_n)\mu_0(dx)}{\sum_{I:\mathcal{A}_N}\int_{A_I} f^s_n(x|\mathbf{y}_n)\mu_0(dx)} = \frac{H^s_{E_J,n}}{H^s_{E_J,n} + H^s_{E^c_J,n} + \sum_{I\neq J} H^s_{I,n}}. \quad (15)$$

That is, we split the $H^s_{A_J,n}$ term in the denominator as an integral over $E_j = E \cap A_J$ plus another over $E^c_J = E^c \cap A_J$. Without loss of generality, assume that $\mu_0(E_J) \neq 0$. Since $0 < F_x\big(B_s(y)\big) \leq 1$ for any $y \in \mathrm{supp}(\tau) \subseteq \mathrm{supp}(F_x)$, and since the contraction maps $s \mapsto F_x\big(B_s(y)\big)$ are locally bi-Lipschitz, the sequence of functions $\{f^s_n\}_{n=1}^\infty$ is pointwise monotonically decreasing to zero with $0 < f^s_n(x) \leq 1$ for all n and any $s > 0$. By the Monotone Convergence Theorem, each sequence $\{H^s_{E,n}\}_{n=1}^\infty$ also decays monotonically to zero with $0 < H^s_{E,n} \leq 1$ for any sequence of data $\{y_k\}_{k=1}^\infty$, any Borel set $E \in \Sigma_X$, and any $s > 0$. Given this, and given that there is an $H^s_{E_J}$ term in both the numerator and denominator of (15), we conclude that $I^s_{N,s}(\mathbf{y}_n|E_J)$ does not have a singularity as $n \to \infty$ and as $s \to 0$. Further, from these considerations, it follows that convergence of $I^s_{N,n}(\mathbf{y}_n|E_J)$ hinges on the relative decay rates of the different $H^s_{E,n}$ terms. Additionally, we must confirm that we can let the s parameter go to zero. We proceed as follows.

Let $\{\varepsilon_m\}_{m=1}^\infty$ be any sequence with $0 < \varepsilon_m < 1$ and $\varepsilon_m \to 0$ as $m \to \infty$. Define

$$P \coloneqq \min\left\{\frac{\partial}{\partial s}_{|s=0}\mathrm{ev}_F(x,s,y) : x \in \mathrm{supp}(\mu_0), y \in \mathrm{supp}(\tau)\right\},$$

and define Q as the maximum over the same set. These values are well-defined, since the supports are compact and the derivatives of the contraction maps at zero vary continuously. Take $\delta_m > 0$ so that

$$\left|\frac{\partial}{\partial s}_{|s=0}\mathrm{ev}_F(x_1,s,y_1) - \frac{\partial}{\partial s}_{|s=0}\mathrm{ev}_F(x_2,s,y_2)\right| < \frac{\varepsilon_m P^2}{3Q} \quad (16)$$

whenever $d^p_X(x_1,x_2) + d^p_Y(y_1,y_2) < \delta^p_m$. Since $\mathrm{ev}_F : X \times [0,1] \times Y \to [0,1]$ is jointly continuous with respect to the product metric on $X \times [0,1] \times Y$, we can additionally take $\sigma_m > 0$ such that

$$\big|F_{x_1}\big(B_s(y_1)\big) - F_{x_2}\big(B_s(y_2)\big)\big| < \varepsilon_m \quad (17)$$

whenever

$$d^p_X(x_1,x_2) + |s-s|^p + d^p_Y(y_1,y_2) = d^p_X(x_1,x_2) + d^p_y(y_1,y_2) < \sigma^p_m.$$

Define $\{\xi_m\}_{m=1}^\infty$ to be any sequence with $\xi_m < \min\left\{\frac{\delta_m}{2^{1/p}}, \frac{\sigma_m}{2^{1/p}}\right\}$. Take $N = N(\xi_m)$ sufficiently large so that $\mathrm{Res}(\mathcal{A}_N)$ and $\mathrm{Res}(\mathcal{C}_N)$ are both at most ξ_m. Then, (16) and (17) hold for any $x_1, x_2 \in A_{j_1,\ldots,j_N} \in \mathcal{A}_N$ and any $y_1, y_2 \in C_{k_1,\ldots,k_N} \in \mathcal{C}_N$, where we are now using the multi-index notation $C_K = C_{k_1,\ldots,k_N}$, rather than the enumeration of $\mathcal{C}_N$ from before. Since the evaluation map ev_F is jointly continuous, and since $[0,1]$ and the closures $\mathrm{Cl}(A_I)$ and $\mathrm{Cl}(C_K)$ of the partition elements are all compact, the maps $\mathrm{Min}_{I,K}(-) : [0,1] \to [0,1]$ and $\mathrm{Max}_{I,K}(-) : [0,1] \to [0,1]$ defined by

$$\mathrm{Min}_{I,K}(s) \coloneqq \min_{x\in\mathrm{Cl}(A_I)}\ \min_{y\in\mathrm{Cl}(C_K)} F_x\big(B_s(y)\big), \text{ and}$$

$$\mathrm{Max}_{I,K}(s) \coloneqq \max_{x\in \mathrm{Cl}(A_I)} \max_{y\in \mathrm{Cl}(C_K)} F_x\big(B_s(y)\big)$$

are well-defined and continuous for any $I : \mathcal{A}_N$ and $K : \mathcal{C}_N$. We also use the notation $P^s_{I,K} \coloneqq \mathrm{Min}_{I,K}(s)$ and $Q^s_{I,K} \coloneqq \mathrm{Max}_{I,K}(s)$, and we additionally write $P^s_{E_J,-}$ and $Q^s_{E_J,-}$ to denote the minimum and maximum values taken over the closure of $E \cap A_J$, rather than A_J alone.

For any data sequence $\{y_k\}_{k=1}^\infty \subseteq C_K$, any $E \in \Sigma_X$, any $I, J : \mathcal{A}_N$, and any $s > 0$, we have

$$\big(P^s_{I,K}\big)^n \le H^s_I(\mathbf{y}_n) \le \big(Q^s_{I,K}\big)^n \quad \text{and} \quad \big(P^s_{E_J,K}\big)^n \le H^s_{E_J}(\mathbf{y}_n) \le \big(Q^s_{E_J,K}\big)^n .$$

Additionally,

$$\frac{\mu_0(E_J)(P^s_{E_J,K})^n}{\sum_{I:\mathcal{A}_N} \mu_0(A_I)(Q^s_{I,K})^n} \le \frac{H^s_{E_J}(\mathbf{y}_n)}{\sum_{I:\mathcal{A}_N} H^s_I(\mathbf{y}_n)} \le \frac{\mu_0(A_J)(Q^s_{E_J})^n}{\sum_{I:\mathcal{A}_N} \mu_0(A_I)(P^s_{I,K})^n}.$$

From this, we can write lower and upper estimates for (13), where we now switch back to our enumerated notation for $\mathcal{C}_N$. Fix $J : \mathcal{A}_N$, $k : [r]$, $L = (\ell_1, \dots, \ell_r)$, and $T(k) = (t_1, \dots, t_{k-1})$ with $\hat{T}(k) \le n-1$. Then, treating F as constant over the partition sets, we divide out everything in the numerators and arrive at sums of geometric series in the denominators:

$$\begin{aligned}
V^{m,s}_{J,k,L,T,n} &\coloneqq \frac{\tau^{\otimes n}\big(C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}\big)}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(\frac{Q^s_{I,\ell_1}}{P^s_{E_J,\ell_1}}\right)^{t_1} \cdots \left(\frac{Q^s_{I,\ell_k}}{P^s_{E_J,\ell_k}}\right)^{n-\hat{T}(k)}} \\
&= \frac{1}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(\frac{Q^s_{I,\ell_1}}{\tau(C_{\ell_1})P^s_{E_J,\ell_1}}\right)^{t_1} \cdots \left(\frac{Q^s_{I,\ell_k}}{\tau(C_{\ell_k})P^s_{E_J,\ell_k}}\right)^{n-\hat{T}(k)}} \\
&\le \int_{C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}} I^s_{N,n}(\mathbf{y}_n|E_J)\tau^{\otimes n}(d\mathbf{y}_n) \\
&\le \frac{1}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(\frac{P^s_{I,\ell_1}}{\tau(C_{\ell_1})\big(Q^s_{E_J,\ell_1}\big)}\right)^{t_1} \cdots \left(\frac{P^s_{I,\ell_k}}{\tau(C_{\ell_k})\big(Q^s_{E_J,\ell_k}\big)}\right)^{n-\hat{T}(k)}} \\
&= \frac{\tau^{\otimes n}\big(C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}\big)}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(\frac{P^s_{I,\ell_1}}{Q^s_{E_J,\ell_1}}\right)^{t_1} \cdots \left(\frac{P^s_{I,\ell_k}}{Q^s_{E_J,\ell_k}}\right)^{n-\hat{T}(k)}} \\
&\eqqcolon W^{m,s}_{J,k,L,T,n}.
\end{aligned} \tag{18}$$

Next we eliminate the s parameter by letting $s \to 0$. By assumption, F has continuously varying contraction maps, so

$$\begin{aligned}
M^0_{I,K} &\coloneqq \min\{M^0_{x,y} : x \in \mathrm{Cl}(A_I), y \in \mathrm{Cl}(C_K)\}, \text{ and} \\
M^1_{I,K} &\coloneqq \max\{M^1_{x,y} : x \in \mathrm{Cl}(A_I), y \in \mathrm{Cl}(C_K)\}
\end{aligned}$$

are well-defined. $M^i_{E_J,K}$ is defined similarly with $\mathrm{Cl}(E_J)$ in place of $\mathrm{Cl}(A_I)$. For any $I, J : \mathcal{A}_N$ and any $K : \mathcal{C}_N$,

$$\frac{M^0_{I,K}}{M^1_{E_J,K}} \leq \frac{\mathrm{Max}_{I,K}(s)}{\mathrm{Min}_{E_J,K}(s)} \leq \frac{M^1_{I,K}}{M^0_{E_J,K}}.$$

Similar bounds hold for the upper approximation terms. Set $\dot{Q}^0_{I,K} := \frac{d}{ds}|_{s=0} Q^s_{I,K}$, and define $\dot{P}^0_{E_J,K}$ and other values similarly. These values are bounded below by the relevant lower Lipschitz constants. By L'Hôpital's rule, the values

$$R^m_{J,I,K} := \lim_{s\to 0} \frac{\mathrm{Max}_{I,K}(s)}{\mathrm{Min}_{E_J,K}(s)} = \lim_{s\to 0} \frac{Q^s_{I,K}}{P^s_{E_J,K}} = \frac{\dot{Q}^0_{I,K}}{\dot{P}^0_{E_J,K}}, \text{ and}$$

$$S^m_{J,I,K} := \lim_{s\to 0} \frac{\mathrm{Min}_{I,K}(s)}{\mathrm{Max}_{E_J,K}(s)} = \lim_{s\to 0} \frac{P^s_{I,K}}{Q^s_{E_J,K}} = \frac{\dot{P}^0_{I,K}}{\dot{Q}^0_{E_J,K}}$$

are defined and nonzero. By continuity, we can take the limit as $s \to 0$ for the entire $V^{m,s}_{J,k,L,T,n}$ term. This gives us

$$\begin{aligned}
V^m_{J,k,L,T,n} &:= \lim_{s\to 0} \frac{\tau^{\otimes n}\big(C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}\big)}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(\frac{Q^s_{I,\ell_1}}{P^s_{E_J,\ell_1}}\right)^{t_1} \cdots \left(\frac{Q^s_{I,\ell_k}}{P^s_{E_J,\ell_k}}\right)^{n-\hat{T}(k)}} \\
&= \frac{\tau^{\otimes n}\big(C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}\big)}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(R^m_{J,I,\ell_1}\right)^{t_1} \cdots \left(R^m_{J,I,\ell_k}\right)^{n-\hat{T}(k)}} \\
&\leq \int_{C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}} \lim_{s\to 0} I^s_{N,n}(\mathbf{y}_n|E_J)\tau^{\otimes n}(d\mathbf{y}_n) \\
&\leq \frac{\tau^{\otimes n}\big(C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}\big)}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(S^m_{J,I,\ell_1}\right)^{t_1} \cdots \left(S^m_{J,I,\ell_k}\right)^{n-\hat{T}(k)}} \\
&= \lim_{s\to 0} \frac{\tau^{\otimes n}\big(C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}\big)}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(\frac{P^s_{I,\ell_1}}{Q^s_{E_J,\ell_1}}\right)^{t_1} \cdots \left(\frac{P^s_{I,\ell_k}}{Q^s_{E_J,\ell_k}}\right)^{n-\hat{T}(k)}} \\
&=: W^m_{J,k,L,T,n}.
\end{aligned}$$

We compute the limit with respect to n as follows:

$$\begin{aligned}
V^m_{J,k,L,T} &:= \lim_{n\to\infty} V^m_{J,k,L,T,n} \qquad (19) \\
&= \lim_{n\to\infty} \frac{1}{1 + \sum_{I:\mathcal{A}'_N} \frac{\mu_0(A_I)}{\mu_0(E_J)} \left(R^m_{J,I,\ell_1}/\tau(C_{\ell_1})\right)^{t_1} \cdots \left(R^m_{J,I,\ell_k}/\tau(C_{\ell_k})\right)^{n-\hat{T}(k)}}
\end{aligned}$$

$$= \begin{cases} 0 & \text{if } \exists I \text{ s.t. } \frac{R^m_{J,I,\ell_k}}{\tau(C_{\ell_k})} > 1 \\ \frac{\tau(C_{\ell_1})^{t_1}\cdots\tau(C_{\ell_{k-1}})^{t_{k-1}}}{1+\sum_{I:\mathcal{A}_N^+}\frac{\mu_0(A_I)}{\mu_0(E_J)}\left(R^m_{J,I,\ell_1}\right)^{t_1}\cdots\left(R^m_{J,I,\ell_{k-1}}\right)^{t_{k-1}}} & \text{if } \forall I, \frac{R^m_{J,I,\ell_k}}{\tau(C_{\ell_k})} \le 1. \end{cases}$$

Here, $\mathcal{A}_N^+$ is the set of those indices where we have equality. A similar result holds for $\mathcal{W}^m$. Now, we have

$$\begin{aligned} \mathcal{V}^m_n &:= \sum_{J:\mathcal{A}_N}\sum_{k:[r]}\sum_{L:\mathcal{L}^r_k}\sum_{T:\mathbb{Z}^k_+} \mathbf{1}_{\{\hat{T}=n\}} V^m_{J,k,L,T,n} \\ &\le \mathbb{E}_{\tau^{\otimes n}}[\mu_n(E)] \\ &\le \sum_{J:\mathcal{A}_N}\sum_{k:[r]}\sum_{L:\mathcal{L}^r_k}\sum_{T:\mathbb{Z}^k_+} \mathbf{1}_{\{\hat{T}=n\}} W^m_{J,k,L,T,n} \\ &=: \mathcal{W}^m_n. \end{aligned}$$

Also, $0 \le \mathcal{V}^m := \lim_{n\to\infty}\mathcal{V}^m_n \le 1$, $0 \le \mathcal{W}^m := \lim_{n\to\infty}\mathcal{W}^m_n \le 1$, and $\mathcal{V}^m \le \mathcal{W}^m$.

Next, we fix n and let $m \to \infty$. Let $I, J : \mathcal{A}_N$, let $K : \mathcal{C}_N$, and take any $N_1 \ge N$. Further, let $I^+, J^+ : \mathcal{A}_{N_1}$ and $K^+ : \mathcal{C}_{N_1}$ be any multi-indices with $A_{I^+} \subseteq A_I$, $E_{J^+} \subseteq E_J$, and $C_{K^+} \subseteq C_K$. From (16),

$$\left|\dot{Q}^0_{I,K} - \dot{Q}^0_{I^+,K^+}\right| < \frac{\varepsilon_m P^2}{3Q} \text{ and } \left|\dot{P}^0_{E_J,K} - \dot{P}^0_{E_{J^+},K^+}\right| < \frac{\varepsilon_m P^2}{3Q}.$$

Set $\omega := \frac{\varepsilon_m P^2}{3Q}$. Since P and Q are nonzero, we may assume that $\omega < P \le \dot{P}^0_{E_J,K}$ and $\omega < Q \le \dot{Q}^0_{I,K}$. Then,

$$\left|\frac{\dot{Q}^0_{I,K}}{\dot{P}^0_{E_J,K}} - \frac{\dot{Q}^0_{J^+,K^+}}{\dot{P}^0_{E_{J^+},K^+}}\right| < \max\left\{\left|\frac{\dot{Q}^0_{I,K}}{\dot{P}^0_{E_J,K}} - \frac{\dot{Q}^0_{I,K}-\omega}{\dot{P}^0_{E_J,K}+\omega}\right|, \left|\frac{\dot{Q}^0_{I,K}}{\dot{P}^0_{E_J,K}} - \frac{\dot{Q}^0_{I,K}+\omega}{\dot{P}^0_{E_J,K}-\omega}\right|\right\}.$$

Further,

$$\left|\frac{\dot{Q}^0_{I,K}}{\dot{P}^0_{E_J,K}} - \frac{\dot{Q}^0_{I,K}-\omega}{\dot{P}^0_{E_J,K}+\omega}\right| \le \frac{\omega\big(\dot{Q}^0_{I,J}+\dot{P}^0_{E_J,K}\big)}{\dot{P}^0_{E_J,K}\big(\dot{P}^0_{E_J,K}+\omega\big)} \le \frac{2\omega Q}{P^2} < \varepsilon_m,$$

and

$$\left|\frac{\dot{Q}^0_{I,K}}{\dot{P}^0_{E_J,K}} - \frac{\dot{Q}^0_{I,K}+\omega}{\dot{P}^0_{E_J,K}-\omega}\right| \le \frac{\omega\big(\dot{Q}^0_{I,J}+\dot{P}^0_{E_J,K}\big)}{\dot{P}^0_{E_J,K}\big(\dot{P}^0_{E_J,K}-\omega\big)} \le \frac{2\omega Q}{P^2-\omega Q}.$$

The above is at most ε_m if and only if $\omega \le \frac{\varepsilon_m P^2}{2Q+\varepsilon_m Q}$, which is indeed the case. Thus, for all $m_1 \ge m$, $\left|R^m_{J,I,K} - R^{m_1}_{J^+,I^+,K^+}\right| < \varepsilon_m$, and a similar analysis shows that $\left|S^m_{J,I,K} - S^{m_1}_{J^+,I^+,K^+}\right| < \varepsilon_m$.

We can now relate $\mathcal{V}^{m_1}_n$ to $\mathcal{V}^m_n$ using the nesting property of the partitions. Any set A_I in $\mathcal{A}_N$ can be written uniquely as a disjoint union of $q = q(I, N_1)$ sets in $\mathcal{A}_{N_1}$:

$$A_I = A_{I,I_1^+} \cup \cdots \cup A_{I,I_q^+}. \tag{20}$$

That is, in contrast to our previous notation, (I, I_j^+) now denotes some multi-index of the form $(i_1, \dots, i_N, i_{N+1}, \dots, i_{N_1})$, and we take exactly those extensions of I giving the equality above. Further, the enumerated sets $C_{K_1}, \dots, C_{K_r}$ of $\mathcal{C}_N$ can be decomposed as

$$C_i = C_{K_i} = C_{K_i, K_{i,1}^+} \cup \cdots \cup C_{K_i, K_{i,q}^+} = C_{i1} \cup \cdots \cup C_{iq}, \tag{21}$$

where $q = q(K_i, N_1)$. This gives us an enumeration of the sets in $\mathcal{C}_{N_1}$ that accounts for the nesting property. That is, we can order the sets in $\mathcal{C}_{N_1}$ as C_{ik}, where $C_{ik} \leq C_{j\ell}$ if $i \leq j$ and $k \leq \ell$. Thus, given $L = (\ell_1, \dots, \ell_k) \in \mathcal{L}_k^r$, we can write $\mathcal{L}(L, k)$ for all $L^+ = (\ell_1', \dots, \ell_k')$ such that $C_{\ell_j, \ell_j'} \subseteq C_{\ell_j}$. Since the partitions are nested, and since the above decomposition is unique, summing over $\mathcal{L}_k^{r_1}$ in level N_1 is equivalent to summing over $L : \mathcal{L}_k^r$ followed by $L^+ : \mathcal{L}(L, k)$. Finally, we write

$$E_J = E \cap A_J = \left(E \cap A_{J, J_1^+}\right) \cup \cdots \cup \left(E \cap A_{J, J_q^+}\right) = E_{J, J_1^+} \cup \cdots \cup E_{J, J_q^+}, \tag{22}$$

where, once again, q will always implicitly depend on the prefixed multi-index as well as the level N_1 of the partition.

Abbreviating E_{J, J_j^+} as E_{J_j}, we can now write

$$V^{m_1}_{(J, J_j^+), k, (L, L^+), T, n} = \frac{\tau^{\otimes n}\left(C^{t_1}_{\ell_1, \ell_1'} \times \cdots \times C^{n - \hat{T}(k)}_{\ell_k, \ell_k'}\right)}{1 + \frac{1}{\mu_0(E_{J_j})}\left(\sum_{I : \mathcal{A}_N'} \sum_{i=1}^{q(I)} \mu_0\left(A_{I, I_i^+}\right) \mathcal{R}^{m_1}_{(J, J_j^+), (I, I_i^+), (L, L^+)}\right)},$$

where

$$\mathcal{R}^{m_1}_{(J, J_j^+), (I, I_i^+), (L, L^+)} \coloneqq \left(R^{m_1}_{(J, J_j^+), (I, I_i^+), (\ell_1, \ell_1')}\right)^{t_1} \cdots \left(R^{m_1}_{(J, J_j^+), (I, I^+), (\ell_k, \ell_k')}\right)^{n - \hat{T}(k)}.$$

From our previous analysis, we know that

$$\left| R^m_{J, I, L} - R^{m_1}_{(J, J_j^+), (I, I_i^+), (L, L^+)} \right| < \varepsilon_m$$

for any suffix extensions of J, I, and L. Thus, we can replace $\mathcal{R}^{m_1}_{(J, J_j^+), (I, I_i^+), (L, L^+)}$ with

$$\mathcal{R}^m_{J, I, L, T}(\varepsilon_m) \coloneqq \left(R^m_{J, I, \ell_1} + \varepsilon_m\right)^{t_1} \cdots \left(R^m_{J, I, \ell_k} + \varepsilon_m\right)^{n - \hat{T}(k)}.$$

Further, set

$$\mathcal{S}_{J_j} \coloneqq \frac{\mu_0(E_{J_1})}{\mu_0(E_{J_j})} + \cdots + \frac{\mu_0(E_{J_j})}{\mu_0(E_{J_j})} + \cdots + \frac{\mu_0(E_{J_q})}{\mu_0(E_{J_j})}$$

and note that

$$\mu_0(E_{J_j}) S_{J_j} = \mu_0(E_{J_1}) + \cdots + \mu_0(E_{J_q}) = \mu_0(E_J).$$

Since $\mathcal{R}^m_{J,I,L}$ is constant for any $i \in [q(I)]$, we use the fact that $\sum_{i=1}^{q(I)} \mu_0\big(A_{I,I_i^+}\big) = \mu_0(A_I)$ to eliminate the sum over suffix extensions $i \in [q(I)]$:

$$
\begin{aligned}
&V^{m_1}_{(J,J_j^+),k,(L,L^+),T,n} \\
&\qquad \geq \frac{\tau^{\otimes n}\big(C^{t_1}_{\ell_1,\ell_1'} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k,\ell_k'}\big)}{\mathcal{S}_{J_j}\mathcal{R}^m_{J,J,L,T}(\varepsilon_m) + \frac{1}{\mu_0(E_{J_j})}\Big(\sum_{I:\mathcal{A}''_N}\sum_{i=1}^{q(I)} \mu_0\big(A_{I,I_i^+}\big)\mathcal{R}^m_{J,I,L,T}(\varepsilon_m)\Big)} \\
&\qquad = \frac{\mu_0(E_{J_j})\tau^{\otimes n}\big(C^{t_1}_{\ell_1,\ell_1'} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k,\ell_k'}\big)}{\mu_0(E_J)\mathcal{R}^m_{J,J,L,T}(\varepsilon_m) + \Big(\sum_{I:\mathcal{A}'_N} \mu_0(A_I)\mathcal{R}^m_{J,I,L,T}(\varepsilon_m)\Big)} \\
&\qquad = \mu_0(E_{J_1})\tau^{\otimes n}\big(C^{t_1}_{\ell_1,\ell_1'} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k,\ell_k'}\big) Z^m_{J,k,L,T,n}(\varepsilon_m),
\end{aligned}
$$

where

$$
Z^m_{J,k,L,T,n}(\varepsilon_m) := \frac{1}{\mu_0(E_J)\mathcal{R}^m_{J,J,L,T}(\varepsilon_m) + \sum_{I:\mathcal{A}'_N} \mu_0(A_I)\mathcal{R}^m_{J,I,L,T}(\varepsilon_m)}.
$$

Appealing once again to the nested property of the partitions and the uniqueness of the decompositions in (20), (21), and (22), we can now eliminate the sums over the other suffix extensions. Setting

$$
\tau^{\otimes n}_{L,L^+,T} := \tau^{\otimes n}\big(C^{t_1}_{\ell_1,\ell_1'} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k,\ell_k'}\big) \ \text{ and } \ \tau^{\otimes n}_{L,T} := \tau^{\otimes n}\big(C^{t_1}_{\ell_1} \times \cdots \times C^{n-\hat{T}(k)}_{\ell_k}\big)
$$

we have

$$
\begin{aligned}
\mathcal{V}^{m_1}_n &= \sum_{J:\mathcal{A}_{N_1}} \sum_{k:[r]} \sum_{L:\mathcal{L}_k^{r_1}} \sum_{T:\mathbb{Z}_+^k} \mathbf{1}_{\{\hat{T}=n\}} V^{m_1}_{J,k,L,T,n} \\
&= \sum_{J:\mathcal{A}_N} \sum_{j:[q(J)]} \sum_{k:[r]} \sum_{L:\mathcal{L}_k^r} \sum_{L^+:\mathcal{L}(L,k)} \sum_{T:\mathbb{Z}_+^k} \mathbf{1}_{\{\hat{T}=n\}} V^{m_1}_{(J,J_j^+),k,(L,L^+),T,n} \\
&\geq \sum_{J:\mathcal{A}_N} \sum_{j:[q(J)]} \sum_{k:[r]} \sum_{L:\mathcal{L}_k^r} \sum_{L^+:\mathcal{L}(L,k)} \sum_{T:\mathbb{Z}_+^k} \mathbf{1}_{\{\hat{T}=n\}} \tau^{\otimes n}_{L,L^+,T}\mu_0(E_{J_j}) Z^m_{J,k,L,T,n}(\varepsilon_m) \\
&= \sum_{J:\mathcal{A}_N} \sum_{j:[q(J)]} \sum_{k:[r]} \sum_{L:\mathcal{L}_k^r} \sum_{T:\mathbb{Z}_+^k} \mathbf{1}_{\{\hat{T}=n\}} \tau^{\otimes n}_{L,T}\mu_0(E_{J_j}) Z^m_{J,k,L,T,n}(\varepsilon_m) \\
&= \sum_{J:\mathcal{A}_N} \sum_{k:[r]} \sum_{L:\mathcal{L}_k^r} \sum_{T:\mathbb{Z}_+^k} \mathbf{1}_{\{\hat{T}=n\}} \tau^{\otimes n}_{L,T}\mu_0(E_J) Z^m_{J,k,L,T,n}(\varepsilon_m) \\
&=: \mathcal{Z}^m_n(\varepsilon_m).
\end{aligned}
$$

By replacing ε_m with $-\varepsilon_m$, and noting that the R terms are bounded away from zero, a similar argument shows that $\mathcal{V}^{m_1}_n \leq \mathcal{Z}^m_n(-\varepsilon_m)$. Thus, for all $m_1 \geq m$, we have $\mathcal{Z}^m_n(\varepsilon_m) \leq \mathcal{V}^{m_1}_n \leq \mathcal{Z}^m_n(-\varepsilon_m)$, and taking the limit with respect to n, as in (19), gives us

$$
\mathcal{Z}^m(\varepsilon_m) \leq \mathcal{V}^{m_1} \leq \mathcal{Z}^m(-\varepsilon_m). \tag{23}
$$

Now, note that

$$|\mathcal{Z}^m(\varepsilon_m) - \mathcal{Z}^m(-\varepsilon_m)| \tag{24}$$
$$= \left| \sum_{J:\mathcal{A}_N} \sum_{k:[r]} \sum_{L:\mathcal{L}_k^r} \sum_{T:\mathbb{Z}_+^k} \tau_{L,T}\mu_0(E_J)\Big(Z^m_{J,k,L,T}(\varepsilon_m) - Z^m_{J,k,L,T}(-\varepsilon_m)\Big)\right|,$$

where $\tau_{L,T} := \tau(C_{\ell_1})^{t_1}\cdots\tau(C_{\ell_{k-1}})^{t_{k-1}}$. After taking common denominators for each difference $Z^m_{J,k,L,T}(\varepsilon_m) - Z^m_{J,k,L,T}(\varepsilon_m)$ and then simplifying, we are left with a sum of terms over J, k, L, and T, each with at least one factor of ε_m in its numerator. We can then evaluate the limit by simple substitution to see that $|\mathcal{Z}^m(\varepsilon_m) - \mathcal{Z}^m(-\varepsilon)|$ goes to zero as $m \to \infty$. Thus, from (23), we have that $\mathcal{V}^m$ is Cauchy in m. That is, we can make $|\mathcal{V}^{m_1} - \mathcal{V}^{m_2}| \le |\mathcal{Z}^m(\varepsilon_m) - \mathcal{Z}^m(-\varepsilon_m)|$ arbitrarily small for all $m_1, m_2 \ge m$ given ε_m sufficiently small. A similar argument to everything above shows that $\mathcal{W}^m$ is also Cauchy in m. Thus, $\mathcal{V} := \lim_{m\to\infty}\mathcal{V}^m$ and $\mathcal{W} := \lim_{m\to\infty}\mathcal{W}^m$ both exist. Since (16) implies $R^m_{J,I,K}$ converges to $S^m_{J,I,K}$ as $m \to \infty$, a term-by-term comparison similar to (24) shows that $\mathcal{V} = \mathcal{W}$. Let $\mathcal{U}$ denote this common value. For any $\varepsilon > 0$, there exists m_0 such that for any $m \ge m_0$, $|\mathcal{V}^m - \mathcal{U}| < \varepsilon/2$ and $|\mathcal{W}^m - \mathcal{U}| < \varepsilon/2$. Further, take $n_0 = n_0(m_0)$ sufficiently large so that $|\mathcal{V}^{m_0}_n - \mathcal{V}^{m_0}| < \varepsilon/2$ and $|\mathcal{W}^{m_0}_n - \mathcal{W}^{m_0}| < \varepsilon/2$ for all $n \ge n_0$. Then, for all $n \ge n_0$,

$$\mathcal{U} - \varepsilon \le \mathcal{V}^{m_0} - \frac{\varepsilon}{2} \le \mathcal{V}^{m_0}_n \le \mathbb{E}_{\tau^{\otimes n}}[\mu_n(E)] \le \mathcal{W}^{m_0}_n \le \mathcal{W}^{m_0} + \frac{\varepsilon}{2} \le \mathcal{U} + \varepsilon.$$

Thus, $\bar{\mu}(E) = \lim_{n\to\infty}\mathbb{E}_{\tau^{\otimes n}}[\mu_n(E)]$ exists (as a real number) and is equal to $\mathcal{U}$.

To see that $\{\bar{\mu}_n\}_{n=1}^\infty$ is Cauchy, again let $\varepsilon > 0$, and take N large enough so that $\mathrm{Res}(\mathcal{A}_N) < \varepsilon/2$. Let $A_1, \ldots, A_D$ be an enumeration of the sets in $\mathcal{A}_N$. Since $\mathrm{supp}(\mu_0)$ is compact, it is bounded. Specifically, there is $B > 0$ such that $\mathrm{supp}(\mu_0)$ is contained in a ball of radius B. Since $\{\mu_n\}_{n=1}^\infty$ converges on any measurable set E, there exists $n(k)$ for each $1 \le k \le D$ such that

$$|\mu_{n_1}(A_k) - \mu_{n_2}(A_k)| < \frac{\varepsilon}{2D(D-1)B^p} =: \varepsilon_0$$

for any $n_1, n_2 \ge n(k)$. Take $n_0 = \max\{n(1), \ldots, n(D)\}$. We can take a coupling $\gamma \in \Pi(\mu_{n_1}, \mu_{n_2})$ that rearranges all but an ϵ_0 amount of mass within the elements of $\mathcal{A}_N$ and then redistributes the remaining mass equally among the remaining sets of the partition. As such,

$$\begin{aligned}
W^p_{p,d_X}(\mu_{n_1}, \mu_{n_2}) &\le \int_{X\times X} d^p_X(x,y)\gamma(dx,dy) \\
&= \sum_{i=1}^{r(N)} \int_{A_i\times A_i} d^p_X(x,y)\gamma(dx,dy) + \sum_{i\ne j}\int_{A_i\times A_j} d^p_x(x,y)\gamma(dx,dy) \\
&\le (\mathrm{Res}(\mathcal{A}_N))^p + \varepsilon_0 D(D-1)B^p \\
&\le \varepsilon.
\end{aligned}$$

Thus, $\{\mu_n\}_{n=1}^{\infty}$ is Cauchy in the Wasserstein space, and as such, the limit exists and must equal $\bar{\mu}$. Thus, $\bar{\mu} \in \mathcal{P}_p(X)$ is a fixed point of $E_X \circ F_\tau : \mathcal{P}_p(X) \to \mathcal{P}_p(X)$. That is,

$$\bar{\mu} = (E_X \circ F_\tau)(\bar{\mu}) = F_{\bar{\mu}}^{\dagger} \odot \tau.$$

□

6. Conclusion and Future Work

In this work, we have laid the foundations for a theory of Bayesian inference using Wasserstein-valued Markov kernels. We showed, in particular, that Markov kernels can be viewed as Borel measurable maps with values in a Wasserstein space via an isometric isomorphism result in Theorem 2.25; we established an altered version of the Disintegration Theorem in Theorem 3.16 using an original theory of integration for Markov kernels developed in Section 3.1; and in Theorem 4.10, we showed that the Kantorovich monad can be defined for Wasserstein metrics of any order. In Section 5, we employed the machinery developed in Sections 2 through 4 to first define Bayesian inverse maps as certain Wasserstein-valued Markov kernels. Further, we used our approximation machinery to give a simplified proof of a generalized Bayes's Law in Theorem 5.4, and we proved two convergence results for the expected Bayesian update in Proposition 5.7 and Theorem 5.10.

6.1. **Future Work.** For future work, one promising direction, as already mentioned, is to develop the Markov process and contraction map approach to convergence in more depth, paying close attention to connections with synthetic Ricci curvature. Additionally, our current results would generalize from expected Bayesian updates to exact updates upon showing a concentration-of-measure result where

$$\tau\left(\left\{y \in Y : W_{p,d_X}\left(\mu_*, F_{\mu_n}^{\dagger}(y)\right) \geq \varepsilon\right\}\right) \to 0$$

under appropriate conditions. Note also that both Wasserstein gradient flows and Wasserstein barycenters can be discussed naturally in our framework, and building connections with these subjects is also a priority.

6.1.1. *Approximate Inference and Stability.* Besides those areas listed above, a central feature of the present work is that it introduces a unified framework for discussing convergence, approximation, and stability properties of Bayesian inference. As such, in future work, we would like to make inroads into the latter two areas.

Roughly, an approximate Bayesian inverse map in our setting is simply any Markov kernel $G : Y \to \mathcal{P}_p(X)$ that is close to the actual inverse map $F_\mu^{\dagger} : Y \to \mathcal{P}_p(X)$ in the Markov-Wasserstein metric. That is, a good approximate inverse should make the quantity

$$\int_Y W_{p,d_X}^p\left(G(y), F_\mu^{\dagger}(y)\right)(F \odot \mu)(dy)$$

small. Supposing a sequence of Bayesian updates, either exact or expected, converges, we may ask: Under what conditions do approximate Bayesian updates also converge to the true distribution? This question is relevant for algorithm design,

since we would ideally like to use surrogate models whenever possible to cut down on computation time.

In [42] and [43], Owhadi et al. investigate stability properties of Bayesian inference and find that inference processes can be unstable in certain continuous settings. Their results largely depend on studying distributions on spaces of distributions, but their metrics of choice are the Prokhorov and TV metrics. It would be worth investigating similar questions in the setting of iterated Wasserstein spaces.

6.1.2. *Generative Models.* Finally, we remark on the potential applicability of our work to generative models. Abstractly, we can think of a generator in a GAN as a map of the form

$$\Theta_G \to L^p_{\mu_Z}(Z, X) \xrightarrow{(-)_\sharp \mu_Z} \mathcal{P}_p(X).$$

Setting $\widehat{\mu}_Z(f) \coloneqq f_\sharp \mu_Z$, we can view a generator as a Markov kernel

$$G \coloneqq \widehat{\mu}_Z \circ f : \Theta_G \to \mathcal{P}_p(X),$$

where $f : \Theta_G \to L^p_{\mu_Z}(Z, X)$ is some parameterized collection of functions. Additionally, encoders can be viewed as maps of the form

$$E \coloneqq \widehat{G(\theta)} \circ h : \Theta_E \to \mathcal{P}_p(Z).$$

Bayesian GANs [47], bidirectional [25], or cycle-consistent, GANs [56], Wasserstein GANs [3], and iterated Wasserstein GANs [26], among other variants, can all potentially be studied effectively in this more general framework. Additionally, we can consider connections with convergence properties of GANs, as studied in, for example, [15], [16], and [36].

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